Math 214: Introduction to Topology
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Chapter 1

Motivation

1.1 Fixed Point Theorems

1.1.1 One dimensional Fixed Point Theorem

Theorem 1.1. *Let $f : [0,1] \to [0,1]$ be continuous, then $\exists x \in [0,1]$, such that $f(x) = x$.*

Proof. Consider $g(x) = f(x) - x$. Since $f(0) \geq 0$, we have $g(0) \geq 0$, and $f(1) \leq 1$, we have $g(0) \leq 0$. Since f is continuous so is g. Hence, by IVT, g has a zero. that is $\exists x_0 : f(x_0) - x_0 = 0$, i.e, $f(x_0) = x_0$. □

1.1.2 Brouwer's Fixed Point Theorem

Theorem 1.2 (Brouwer's Fixed Point Theorem)**.** *Every continuous map $f : D \to D$ on a closed disk has a fixed point*

1.2 Euler's Identity

Theorem 1.3 (Euler's Identity)**.** *Consider a plane and a graph on the plane containing vertices V, edges E and faces F, this graph satisfies $V - E + F = 2$*

1.3 Euclidean Spaces

Let $X = \mathbb{R}^d$, we say "$x_1, x_2, \ldots$ converges to x" if $\forall \varepsilon > 0, \exists n_0 > 0 : \forall n \geq n_0 : d(x_n, x) < \varepsilon$, i.e, $x_n \in B_\varepsilon(x) \iff lim_{n\to\infty} = x$.
Let $f : \mathbb{R}^d \to \mathbb{R}^d$, be a function, we say f is continuous at every $x \in \mathbb{R}^d$, which means $\forall \epsilon > 0, \exists \delta > 0 : \forall x' \in \mathbb{R}^d, d(x, x') < \delta \to d(f(x), f(x')) < \epsilon$ (i.e., $x' \in B_\delta(x), f(x') \in B_\epsilon(f(x))$, or $f(B_\delta(x)) \subseteq B_\epsilon(f(x))$). If $x_n \to x$ as $n \to \infty$ then $f(x_n) \to f(x)$ as $n \to \infty$

Chapter 2

Introduction

2.1 Axioms of Topology, Closure, Interior

Let $X = \mathbb{R}^d : x_n \to x$, $f : X \to X$ continuous then $\forall x \in X, \forall \epsilon > 0 \exists \delta > 0 : f(B_\delta(x) \subseteq B_\epsilon(f(x))$

Definition 2.1. A set $U \subseteq X$ is said to be open if $\forall x \in U \exists \epsilon > 0 : B_\epsilon(x) \subseteq U$ A set $V \subseteq X$ is said to be closed if its complement is open.

Definition 2.2. We define the closure and the interior of a set as follows:

- Closure $A \to cl(A) = x \in \mathbb{R}^d : \forall \epsilon > 0 B_\epsilon(x) \cap A \neq \phi$
- Interior $A \to int(A) = x \in \mathbb{R}^d : \exists \epsilon > 0 : B_\epsilon(x) \subseteq A$

Definition 2.3. A topology of a set X can be described by specifying which of its subsets are open. The family of open sets $\mathcal{T} \subseteq 2^X$

Definition 2.4 (Axioms of Topology)**.** .

(1) The arbitrary union of open sets is open

(2) The finite intersection of closed sets is closed

(3) $\emptyset, X \in \mathcal{T}$

Example 2.5. .

(1) The trivial Topology: $\emptyset \neq X$ arbitrary, $\mathcal{T} = \{\emptyset, X\}$

(2) The discrete Topology: $\emptyset \neq X$ arbitrary, $\mathcal{T} = 2^X$

Definition 2.6. A topological space is a set with a family of open sets satisfying the axioms we denote it $(X, \mathcal{T})$

Definition 2.7. A sequence $x_1, x_2. \cdots \in X$ converges to a point $x \in X$, if $\forall$ open sets $U : x \in U \quad \exists n_0 : \forall n \geq n_0, x_n \in U$

Definition 2.8. Let $(X, \mathcal{T}), (Y, \Gamma)$ be topological spaces, $f : X \to Y$ is continuous if $\forall U \in \Gamma : f^{-1}(U) \in \mathcal{T}$

Exercise 2.9. $f : \mathbb{R}^d \to X : (X, \mathcal{T})$ is a discrete topology. Show that f is continuous $\iff$ f is constant.

Example 2.10. .

(1) $X = \{0, 1\}$, $\mathcal{T} = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$

(2) Co-finite topology: $\emptyset \neq X$ arbitrary $A \subseteq X$ is open $\iff$ either $X \setminus A$ is finite or $A = \emptyset$

Chapter 3

Bases

Definition 3.1. A topological basis on a set X is a family $\mathcal{B} \subseteq 2^X$ such that $\bigcup_{A\in\mathcal{B}} A = X$. If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then $\exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The topology generated by a basis $\mathcal{B}$ is $\mathcal{T} : \mathcal{T}(b) = \{\cup_{A\in\mathcal{B}} A : \Gamma \subseteq B\} = \{U \subseteq X; \forall x \in U, \exists A \in \mathcal{B} : x \in A \subseteq U\}$

Definition 3.2. A *topological basis* on a set X is a family $\mathcal{B} \subseteq 2^X$. The topology $\mathcal{T}(\mathcal{B})$ *generated by* $\mathcal{B}$ is

$$\mathcal{T}(\mathcal{B}) = \left\{ \bigcup_{A\in\Gamma} A : \Gamma \subseteq \mathcal{B} \right\} = \{\, U \subseteq X : \forall x \in U,\ \exists A \in \mathcal{B} \text{ such that } x \in A \subseteq U \,\}.$$

Proposition 3.3. *If $\mathcal{B}$ is a basis, then $\mathcal{T}(\mathcal{B})$ is a topology.*

Proposition 3.4. *Let $\mathcal{B}_1, \mathcal{B}_2$ be two bases on a set X. Then*

$$\mathcal{T}(\mathcal{B}_1) \subseteq \mathcal{T}(\mathcal{B}_2) \quad \Longleftrightarrow \quad \forall x \in X\, \forall A_1 \in \mathcal{B}_1 \text{ with } x \in A_1,\ \exists A_2 \in \mathcal{B}_2 \text{ such that } x \in A_2 \subseteq A_1.$$

In this case we say that $\mathcal{T}(\mathcal{B}_2)$ is finer *than $\mathcal{T}(\mathcal{B}_1)$.*

Proposition 3.5. *Let $\mathcal{B} \subseteq 2^X$ and define*

$$\mathcal{T} = \left\{ \bigcup_{A\in\Gamma} A : \Gamma \subseteq \mathcal{B} \right\}.$$

If $\mathcal{T}$ is a topology, then $\mathcal{B}$ is a basis. That is:

(1) $\bigcup_{A\in\mathcal{B}} A = X$.

(2) For all $A_1, A_2 \in \mathcal{B}$ and all $x \in A_1 \cap A_2$, there exists $A_3 \in \mathcal{B}$ such that

$$x \in A_3 \subseteq A_1 \cap A_2.$$

Example 3.6. Let $X = \mathbb{R}^2$.

(1) $\mathcal{B}_1$ = the family of open balls; $\mathcal{T}(\mathcal{B}_1)$ is the standard Euclidean topology.

(2) $\mathcal{B}_2$ = the family of open rectangles; $\mathcal{T}(\mathcal{B}_2)$ is also the Euclidean topology.

Chapter 4

Order Topology

4.1 Lexicographic Order

Definition 4.1. The *lexicographic order* on $\mathbb{R}^2$ is the total order defined by

$$(a, b) < (c, d) \quad \text{iff} \quad a < c \text{ or } (a = c \text{ and } b < d).$$

For $a, b \in \mathbb{R}^2$, the corresponding open interval is

$$(a, b) = \{\, x \in \mathbb{R}^2 : a < x < b \,\}.$$

4.2 Order Topology

Definition 4.2. Let $(X, \leq)$ be a totally ordered set. The *order topology* on X is the topology generated by the basis

$\mathcal{B} = \{\, (a, b) : a, b \in X,\, a < b \,\} \cup \{\, [m, b) : b \in X$ and m is a minimum of $X \,\} \cup \{\, (a, M] : a \in X$ and M is a maxim

4.3 Relations Between Topologies on $\mathbb{R}^d$

Proposition 4.3. *On $\mathbb{R}^d$ the following chain of topologies holds:*

$$\textit{Discrete topology} \supseteq \textit{Euclidean topology} \supseteq \textit{Co-finite topology} \supseteq \textit{Trivial topology.}$$

The lexicographic topology on $\mathbb{R}^2$ is not comparable with either the Euclidean topology or the co-finite topology, but it is:

$$\textit{finer than the trivial topology,} \qquad \textit{coarser than the discrete topology.}$$

4.4 Intersection of Topologies

Remark 4.4. Let $\mathcal{T}_i$ be topologies on X. Then

$$\bigcap_{i \in I} \mathcal{T}_i$$

is again a topology.

If $\mathcal{B} \subseteq 2^X$ is a basis, then

$$\mathcal{T}(\mathcal{B}) = \left\{ \bigcup_{A \in \Gamma} A : \Gamma \subseteq \mathcal{B} \right\}.$$

If $\Gamma \subseteq 2^X$ is arbitrary, then the topology generated by Γ is

$$\mathcal{T}(\Gamma) = \bigcap \left\{ \widehat{\mathcal{T}} : \widehat{\mathcal{T}} \text{ is a topology on } X \text{ and } \Gamma \subseteq \widehat{\mathcal{T}} \right\}.$$

4.5 Equivalence of the Two Definitions of $\mathcal{T}(\mathcal{B})$

Lemma 4.5. *If $\mathcal{B} \subseteq 2^X$ is a basis, then the two definitions of $\mathcal{T}(\mathcal{B})$ agree.*

Proof. Let

$$T_1 = \{\text{unions of elements of } \mathcal{B}\}, \qquad T_2 = \text{the coarsest topology containing } \mathcal{B}.$$

Clearly $T_2 \subseteq T_1$, since T_2 is the smallest topology containing $\mathcal{B}$.

For the reverse inclusion, let $U = \bigcup_{A \in \Gamma} A$ with $\Gamma \subseteq \mathcal{B}$. Since each $A \in \Gamma$ is open in T_2, and unions of open sets are open, we have $U \in T_2$. Thus $T_1 \subseteq T_2$. □

4.6 Finite Unions Basis

Lemma 4.6. *Let $\Gamma \subseteq 2^X$ be arbitrary. Define*

$$\mathcal{B}(\Gamma) = \{A_1 \cup A_2 \cup \cdots \cup A_n : n \geq 1 \text{ and } A_i \in \Gamma\}.$$

Then $\mathcal{B}(\Gamma)$ is a basis.

Example 4.7. For a totally ordered set X:

$$(a, b) = \{x \in X : a < x < b\},$$
$$[a, b) = \{x \in X : a \leq x < b\},$$
$$(a, b] = \{x \in X : a < x \leq b\}.$$

4.7 Characterizing the Topology Generated by a Subbasis

Definition 4.8. A family $\Gamma \subseteq 2^X$ is called a *sub-basis* if

$$\bigcup_{A \in \Gamma} A = X.$$

Proposition 4.9. *Let $\Gamma \subseteq 2^X$ be arbitrary. Then:*

(1) $\mathcal{T}(\Gamma) = \mathcal{T}\big(B(\Gamma)\big)$.

(2) In particular,

$$\mathcal{T}(\Gamma) = \{\textit{arbitrary unions of finite intersections of elements of } \Gamma\}.$$

Chapter 5

Product Topology, Convergence, and Continuity

5.1 Product Topology

Definition 5.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. The *product topology* on $X \times Y$ is the topology generated by the basis

$$\mathcal{B} = \{\, U \times V : U \in \mathcal{T}_X,\ V \in \mathcal{T}_Y \,\}.$$

We write $\mathcal{T}(\mathcal{B})$ for the product topology.

Proposition 5.2. *Let $\mathcal{B}_X$ be a basis for $(X, \mathcal{T}_X)$ and $\mathcal{B}_Y$ a basis for $(Y, \mathcal{T}_Y)$. Then*

$$\mathcal{B}_{X\times Y} = \{\, A \times B : A \in \mathcal{B}_X,\ B \in \mathcal{B}_Y \,\}$$

is a basis for the product topology.

Proposition 5.3. *The collection*

$$\Gamma = \{U \times Y : U \in \mathcal{T}_X\} \cup \{X \times V : V \in \mathcal{T}_Y\}$$

is a sub-basis generating the product topology.

5.2 Convergence via Basis or Sub-basis

Proposition 5.4. *Let $(X, \mathcal{T})$ be a topological space, and let $x_n \to x$. Then:*

(1) If $\mathcal{B}$ is a basis for $\mathcal{T}$, then

$$x_n \to x \quad \Longleftrightarrow \quad \forall A \in \mathcal{B} \text{ with } x \in A,\ \exists n_0 \text{ such that } \forall n \geq n_0,\ x_n \in A.$$

(2) If $\mathcal{S}$ is a sub-basis for $\mathcal{T}$, then

$$x_n \to x \quad \Longleftrightarrow \quad \forall A \in \mathcal{S} \text{ with } x \in A,\ \exists n_0 \text{ such that } \forall n \geq n_0,\ x_n \in A.$$

5.3 Continuity via Basis or Sub-basis

Proposition 5.5. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and $f : X \to Y$.*

(1) Suppose $\mathcal{B}_Y$ is a basis for $\mathcal{T}_Y$. Then

$$f \text{ is continuous} \iff \forall B \in \mathcal{B}_Y,\ f^{-1}(B) \in \mathcal{T}_X.$$

(2) Suppose $\mathcal{S}_Y$ is a sub-basis for $\mathcal{T}_Y$. Then

$$f \text{ is continuous} \iff \forall A \in \mathcal{S}_Y,\ f^{-1}(A) \in \mathcal{T}_X.$$

5.4 Subspaces

Definition 5.6. Let $(X, \mathcal{T})$ be a topological space and let $Y \subseteq X$. The *subspace topology* on Y is

$$\mathcal{T}_Y = \{\, A \cap Y : A \in \mathcal{T} \,\}.$$

We call $(Y, \mathcal{T}_Y)$ a *subspace* or *relative topology.*

5.5 Examples

Example 5.7. (1) $X = \mathbb{R}$, subspaces:

$$Y = [a, b], \qquad (a, b), \qquad [a, b).$$

(2) $X = \mathbb{R}^2$,

$$Y = [a, b] \times [c, d], \qquad S' = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

(3) $X = \mathbb{R}^{d+1}$,

$$Y = S^d = \{x \in \mathbb{R}^{d+1} : \|x\| = 1\}.$$

5.6 Product as a Subspace Topology

Proposition 5.8. *Let X and Y be topological spaces. Then the following two topologies on $X \times Y$ coincide:*

(1) the product topology on $X \times Y$;

(2) the subspace topology inherited from the product

$$X' \times Y' \subseteq X \times Y,$$

where X' and Y' have the subspace topologies.

Chapter 6

Closure, Interior, Boundary

6.1 Neighborhoods

Definition 6.1. Let X be a topological space and $x \in X$. An *open neighborhood* of x is an open set U such that $x \in U$.

6.2 Closure

Definition 6.2. Let $E \subseteq X$. The *closure* of E is

$$\mathrm{cl}(E) = \{x \in X : \forall \text{ open neighborhoods } N \ni x,\ N \cap E \neq \varnothing\}.$$

Equivalently:

$$\mathrm{cl}(E) = \bigcap\{\ C \subseteq X : C \text{ is closed and } E \subseteq C\ \},$$

i.e. the smallest closed set containing E.

6.3 Interior

Definition 6.3. Let $E \subseteq X$. The *interior* of E is

$$\mathrm{int}(E) = \{x \in X : \exists \text{ open neighborhood } N \ni x,\ N \subseteq E\}.$$

Equivalently:

$$\mathrm{int}(E) = \bigcup\{\ U : U \text{ open and } U \subseteq E\ \},$$

the largest open set contained in E.

6.4 Boundary

Definition 6.4. For $E \subseteq X$, the *boundary* of E is

$$\partial E = \mathrm{cl}(E) \setminus \mathrm{int}(E) = \mathrm{cl}(E) \cap \mathrm{cl}(X \setminus E).$$

Equivalently:

$$x \in \partial E \quad \Longleftrightarrow \quad \forall \text{ open neighborhoods } N \ni x,\ N \cap E \neq \varnothing \text{ and } N \cap (X \setminus E) \neq \varnothing.$$

6.5 Basic Properties

Proposition 6.5. *For any $E, A, B \subseteq X$:*

(1) $\operatorname{int}(E) \subseteq E \subseteq \operatorname{cl}(E)$.

(2) $\operatorname{cl}(\operatorname{cl}(E)) = \operatorname{cl}(E)$.

(3) If $A \subseteq B$, then $\operatorname{cl}(A) \subseteq \operatorname{cl}(B)$.

(4) $\operatorname{cl}(A \cup B) = \operatorname{cl}(A) \cup \operatorname{cl}(B)$.

(5) $\operatorname{cl}(A \cap B) \subseteq \operatorname{cl}(A) \cap \operatorname{cl}(B)$.

(6) $\operatorname{cl}(\varnothing) = \varnothing$.

(7) E is closed iff $\operatorname{cl}(E) = E$.

(8) $X \setminus \operatorname{int}(E) = \operatorname{cl}(X \setminus E)$.

(9) $X \setminus \operatorname{cl}(E) = \operatorname{int}(X \setminus E)$.

(10) $\operatorname{cl}\Big(\bigcup_{i=1}^{\infty} A_i\Big) \neq \bigcup_{i=1}^{\infty} \operatorname{cl}(A_i)$ *in general.*

Proposition 6.6. *For any $E \subseteq X$:*

(1) $\operatorname{int}(E) \cap \partial E = \varnothing$.

(2) $\partial E = \varnothing$ iff E is clopen.

(3) $\operatorname{cl}(E) = E \cup \partial E = \operatorname{int}(E) \cup \partial E$.

6.6 Isolated and Limit Points

Definition 6.7. Let $E \subseteq X$.

(1) A point $x \in E$ is an *isolated point* of E if

$$\exists \text{ neighborhood } N \ni x \text{ such that } N \cap E = \{x\}.$$

(2) A point $x \in X$ is a *limit point* (accumulation point) of E if

$$\forall \text{ neighborhoods } N \ni x,\ (N \setminus \{x\}) \cap E \neq \varnothing.$$

Definition 6.8. We write:

$$\lim(E) = \{\text{limit points of } E\}, \qquad \operatorname{iso}(E) = \{\text{isolated points of } E\}.$$

Proposition 6.9. *For any $E \subseteq X$:*

$$\operatorname{cl}(E) = \operatorname{iso}(E) \cup \lim(E) = E \cup \lim(E).$$

Proposition 6.10. *For any $E \subseteq X$:*

$$\partial(\partial E) \subseteq \partial E, \qquad \partial E = \partial(X \setminus E).$$

Proposition 6.11. *(1) If x is the limit of a sequence in E, then $x \in \operatorname{cl}(E)$.*

(2) If x is the limit of a sequence in $X \setminus E$, then $x \in \lim(E)$.

6.7 Continuity via Neighborhoods, Closure, and Preimages

Proposition 6.12. *Let $f : X \to Y$ be a map between topological spaces. The following are equivalent:*

(1) For every open $U \subseteq Y$, $f^{-1}(U)$ is open in X.

(2) For every closed $V \subseteq Y$, $f^{-1}(V)$ is closed in X.

(3) For all $E \subseteq X$,

$$f(\mathrm{cl}(E)) \subseteq \mathrm{cl}(f(E)).$$

(4) f is continuous at every $x \in X$: for every neighborhood B of $f(x)$ there exists a neighborhood A of x such that $f(A) \subseteq B$.

Proof of $2 \Rightarrow 3$. Let $V = \mathrm{cl}(f(E))$. By (2), $f^{-1}(V)$ is closed in X and contains E, so

$$\mathrm{cl}(E) \subseteq f^{-1}(V).$$

Applying f,

$$f(\mathrm{cl}(E)) \subseteq f(f^{-1}(V)) \subseteq V = \mathrm{cl}(f(E)).$$

□

Proof of $3 \Rightarrow 2$. Let $V \subseteq Y$ be closed, and set $U = f^{-1}(V)$. Applying (3),

$$f(\mathrm{cl}(U)) \subseteq \mathrm{cl}(f(U)) = \mathrm{cl}(V) = V.$$

Thus $\mathrm{cl}(U) \subseteq f^{-1}(V) = U$, so U is closed. □

6.8 Sequential Continuity

Proposition 6.13. *Let $f : X \to Y$ be continuous and let $x_n \to x$ in X. Then $f(x_n) \to f(x)$.*

Chapter 7

Homeomorphisms and Hausdorff Spaces

7.1 Homeomorphisms

Definition 7.1. A map $f : X \to Y$ between topological spaces is called a *homeomorphism* if:

(1) f is a bijection,

(2) f is continuous,

(3) f^{-1} is continuous.

If such an f exists, we say that X and Y are *homeomorphic*.

Definition 7.2. Two topological spaces X and Y are called *homeomorphic* (or *topologically equivalent*) if there exists a homeomorphism $f : X \to Y$.

7.2 Hausdorff Spaces

Definition 7.3. A topological space X is called *Hausdorff* if for every pair of distinct points $a, b \in X$ there exist neighborhoods $A \ni a$ and $B \ni b$ such that

$$A \cap B = \varnothing.$$

7.3 Features of Hausdorff Spaces

Proposition 7.4. *If X is a Hausdorff space, then:*

(1) Every singleton $\{x\}$ is closed.

(2) Every finite subset of X is closed.

(3) Every convergent sequence has at most one limit.

7.4 Embeddings

Definition 7.5. Let X and Y be topological spaces. An injective continuous map $f : X \to Y$ is called an *embedding* if the inverse map

$$f^{-1} : f(X) \longrightarrow X,$$

where $f(X)$ is equipped with the subspace topology, is continuous. Equivalently, $f : X \to f(X)$ is a homeomorphism.

Chapter 8

Topological Properties, More on Continuity, Urysohn's Lemma, Tietze Extension

8.1 Topological Properties (Invariants)

Definition 8.1. A property P of topological spaces is called a *topological property* or *topological invariant* if it is preserved by homeomorphisms. That is, if $X \cong Y$ and X has property P, then Y also has P.

Example 8.2. Common topological invariants include:

(1) Cardinality,

(2) Hausdorffness,

(3) Connectedness and path connectedness,

(4) Compactness,

(5) Metrizability,

(6) Fundamental group,

(7) Topological dimension.

8.2 More on Continuity

Proposition 8.3. *Basic facts about continuous maps:*

(1) All constant functions are continuous.

(2) The inclusion map $i : X_0 \hookrightarrow X$, $i(x) = x$, is continuous.

(3) If $f : X \to Y$ and $g : Y \to Z$ are continuous, then $g \circ f$ is continuous.

(4) If $f : X \to Y$ is continuous and $X_0 \subseteq X$, then the restriction $f|_{X_0} : X_0 \to Y$ is continuous.

(5) Projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are continuous with respect to the product topology.

Proposition 8.4 (Pasting Lemma)**.** *Let X be a topological space and $A, B \subseteq X$. Assume that A and B are open, or that A and B are closed. Let $f : A \to Y$ and $g : B \to Y$ be continuous, and assume*

$$f|_{A\cap B} = g|_{A\cap B}.$$

Then the map $h : A \cup B \to Y$ defined by

$$h(x) = \begin{cases} f(x), & x \in A, \\ g(x), & x \in B \end{cases}$$

is continuous.

Remark 8.5. The pasting lemma does *not* hold for arbitrary subsets. Example: the Dirichlet function (continuous on rationals and irrationals separately).

8.3 Urysohn's Lemma

Definition 8.6 (Informal Formulation)**.** Given a topological space X and two disjoint subsets $A, B \subseteq X$, the goal of Urysohn's lemma is to construct a continuous function

$$f : X \to [0, 1]$$

such that

$$f(A) = \{0\}, \qquad f(B) = \{1\}.$$

(Full formal statement requires X to be normal.)

8.4 Tietze Extension Theorem

Definition 8.7 (Informal Formulation)**.** Let X be a topological space, let $A \subseteq X$ be closed, and let

$$f : A \to \mathbb{R}$$

be continuous (with the subspace topology). The *Tietze Extension Theorem* asks:

Does there exist a continuous extension

$$g : X \to \mathbb{R} \qquad \text{such that} \qquad g|_A = f \ ?$$

Remark 8.8. A related question: If $f : \mathbb{Q} \to \mathbb{R}$ is continuous (with $\mathbb{Q}$ dense in $\mathbb{R}$), when can f be extended continuously to a map on all of $\mathbb{R}$? (The answer depends on uniform continuity and boundedness.)

Chapter 9

Metrizable Spaces

9.1 Equivalent Metrics

Definition 9.1 (Topological Equivalence of Metrics)**.** Two metrics d_1, d_2 on X are *topologically equivalent* if they generate the same topology.

Remark 9.2. Every metric is equivalent to a bounded one. This is useful when comparing product and uniform metrics.

Example 9.3. (1) $\overline{d}(x,y) = \min\{d(x,y), 1\} \leq 1$.

(2) $\overline{d}(x,y) = \dfrac{d(x,y)}{1+d(x,y)} < 1$.

9.2 Metrizable vs Metric Spaces

Remark 9.4. A metric space $\neq$ a metrizable space: - Every metric space is metrizable (by definition). - But the converse is not tautological: a topology is metrizable if there *exists* a compatible metric.

Example 9.5 (Discrete Topology)**.** On any set X, the discrete topology is metrizable by the discrete metric:

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Example 9.6 (Cantor Product $\{0,1\}^{\mathbb{N}}$)**.** With the product topology,

$$d(x,y) = \begin{cases} 0, & x = y, \\ 2^{-n_b(x,y)}, & x \neq y, \end{cases}$$

where $n_b(x,y) = \min\{n : x_n \neq y_n\}$.

Equivalently,

$$d(x,y) = \sum_{n=1}^{\infty} \frac{d_0(x_n, y_n)}{2^n}, \qquad d_0 = \text{discrete metric on } \{0,1\}.$$

9.3 A Non-Metrizable Example

Example 9.7. Let $\mathcal{X} = \{0,1\}^I$ with product topology where I is uncountable. Then $\mathcal{X}$ is *not metrizable*.

Original Proof. Suppose $\mathcal{X}$ has a metric d. In any metric space,

$$\bigcap_{n=1}^{\infty} B_{2^{-n}}(x) = \{x\}.$$

But in $\mathcal{X}$, basic open neighborhoods are cylinder sets

$$C(K, w) = \{y \in \mathcal{X} : \forall i \in K,\ y_i = w_i\}, \quad K \subset I \text{ finite.}$$

If $\mathcal{X}$ were metrizable, $\{B_{2^{-n}}(x)\}$ would form a countable neighborhood base. Then there exist finite K_n with

$$x \in C(K_n, x|_{K_n}) \subseteq B_{2^{-n}}(x).$$

Thus

$$\bigcap_{n} C(K_n, x|_{K_n}) \subseteq \{x\}.$$

But $\bigcup_n K_n$ is countable, while I is uncountable, so pick $j \in I \setminus \bigcup_n K_n$. Then x and y differing only in coordinate j both lie in the intersection — contradiction. □

Remark 9.8. This shows: a countable base at each point is necessary for metrizability (second countability).

9.4 Topologies on $\mathbb{R}^{\mathbb{N}}$

Example 9.9 (Three Standard Topologies on $\mathbb{R}^{\mathbb{N}}$)**.** (1) **Product topology**: metrizable with

$$d(\underline{x}, \underline{y}) = \sum_{i=1}^{\infty} \frac{\overline{d}(x_i, y_i)}{2^i}.$$

(2) **Uniform topology**: metrizable with

$$d(\underline{x}, \underline{y}) = \sup_{i \in \mathbb{N}} \overline{d}(x_i, y_i).$$

(3) **Box topology**: not metrizable in general.
Relationship:

$$\mathcal{T}_{\text{product}} \subseteq \mathcal{T}_{\text{uniform}} \subseteq \mathcal{T}_{\text{box}}.$$

9.5 Characterization Theorem

Theorem 9.10. *Let $(X_i)_{i \in I}$ be non-trivial metrizable spaces. Then*

$$\prod_{i \in I} X_i \text{ is metrizable} \iff I \text{ is countable.}$$

Remark 9.11. For uncountable products, the product topology is not metrizable. But countable products of metric spaces remain metrizable, as shown above with $\mathbb{R}^{\mathbb{N}}$.

9.6 Sequential Characterizations

Remark 9.12. In metrizable spaces, all topological notions can be described by sequences. This is not true in general topological spaces.

Example 9.13. (1) $\mathrm{cl}(E) = \{x : \exists (x_n) \subseteq E, x_n \to x\}$ (true in metrizable spaces).

(2) $\lim(E) = \{x : \exists (x_n) \subseteq E \setminus \{x\}, x_n \to x\}$.

(3) If $f : X \to Y$ and X is metrizable, then f continuous $\iff x_n \to x \implies f(x_n) \to f(x)$.

Theorem 9.14. *Let X be a metrizable topological space, Y any topological space, and $f : X \to Y$. Then for any $x \in X$,*

$$f \text{ is continuous at } x \quad \iff \quad \big(\forall\, (x_n) \subset X,\ x_n \to x \implies f(x_n) \to f(x)\big).$$

Proof. ($\Rightarrow$) If f is continuous at x and $x_n \to x$ in X, then for every neighborhood V of $f(x)$, $U := f^{-1}(V)$ is a neighborhood of x. Hence $x_n \in U$ eventually, i.e. $f(x_n) \in V$ eventually, so $f(x_n) \to f(x)$.

($\Leftarrow$) Suppose f is not continuous at x. Then there exists an open neighborhood V of $f(x)$ in Y such that for every neighborhood U of x in X, we have $U \not\subseteq f^{-1}(V)$. Using the metric d that induces τ_X, this means:

$$\forall r > 0\ \exists y \in B_{2^{-n}}(x, r) \ \text{ with } \ f(y) \notin V.$$

For each $n \in \mathbb{N}$ choose $x_n \in B_{2^{-n}}\big(x, \frac{1}{n}\big)$ with $f(x_n) \notin V$. Then

$$d(x_n, x) < \tfrac{1}{n} \ \longrightarrow\ 0,$$

so $x_n \to x$ in the metric space (X, d), hence in the topology of X. However, V is a neighborhood of $f(x)$ and $f(x_n) \notin V$ for all n, so $f(x_n) \not\to f(x)$. This contradicts the assumed sequential condition. Therefore f is continuous at x. □

Theorem 9.15. *Let X be a metrizable topological space. Then for any $E \subseteq X$,*

$$\overline{E} = \{\, x \in X : \exists (x_n)_{n \geq 1} \subseteq E \text{ with } x_n \to x \,\}.$$

Proof. ($\supseteq$) True in general.

($\subseteq$) Fix a metric d on X. Let $x \in \overline{E}$, i.e., for every open neighborhood N of x we have $N \cap E \neq \varnothing$. In particular, for each $n \in \mathbb{N}$ the open ball $B_{1/n}(x) = \{y \in X : d(x, y) < 1/n\}$ satisfies $B_{1/n}(x) \cap E \neq \varnothing$. Choose

$$x_n \in B_{1/n}(x) \cap E \quad (n \in \mathbb{N}).$$

Then $d(x_n, x) < \frac{1}{n}$ for all n, hence $x_n \to x$. Thus x is a limit of a sequence from E. □

Definition 9.16. for $x \in X$ is a family $N_i : i \in I$ of neighborhoods of x such that, $\forall$ open $U \ni x, \exists i \in I, N_i \subseteq U$

Definition 9.17. X is first-countable if around each $x \in X$, there exists a countable neighborhood basis.

Chapter 10

Uniform Continuity

Definition 10.1. Let X, Y be metric spaces. A map $f : X \to Y$ is uniformly continuous if:
$\forall \epsilon > 0 \exists \delta > 0$ such that $\forall x, x' \in X$ we have $d(x, x') < \delta \implies d(f(x), f(x')) < \epsilon$

Example 10.2. $f : (0, 1) \to \mathbb{R}$, $f(x) = \frac{1}{x}$. f is continuous but not uniformly.
Let $\epsilon = 1$ and $\delta > 0$ let $x = \delta$ and $x' < \delta$ such that $\frac{1}{x'} > \frac{1}{x}, x' < \frac{x}{1+x} < x$

10.1 Convergence of functions:

Let X be a set and Y a topological space. Two notions of convergence of maps $X \to Y$:

(1) Pointwise Convergence:
$f_n \to f \iff \forall x \in X, f_n(x) \to f(x)$ (convergence in Product topology)

(2) Uniform Convergence (If Y is metric):
$f_n \to f \iff \forall \epsilon > 0, \exists n_0, \forall n \geq n_0, \forall x \in X, |f_n(x) - f(x)| < \epsilon$
This is the notion of convergence with respect to the uniform metric on Y^X.
$p(f, g) = sup\{d(f(x), g(x)) : x \in X\}$

Example 10.3. $X = [0, 1], Y = [0, 1]$,
$f_n(x) = x^n$ then,

$$f(x) = \begin{cases} 1 & x = 1 \\ 0 & \text{otherwise} \end{cases}$$

$f_n \to f$ pointwise but not uniformly

Theorem 10.4 (Uniform Convergence Theorem)**.** *Let X be a topological space and Y be a metric space. Let $f_n : X \to Y$ $(n = 1, 2, 3, \dots)$ be a sequence of continuous functions. If $f_n \to f$ uniformly, then f is continuous.*

There are intepretations to the above theorem:

(1) $C(X, Y) \subseteq Y^X$ is closed with respect to the uniform topology.

(2) Let $f : X \to Y$. If $\epsilon > 0$, $\exists$ continuous $f_\epsilon : X \to Y$ that is ϵ-approximates f, then f is continuous $\forall x \in X$, $d(f(x), f_\epsilon(x)) < \epsilon$

Proof. Let $x_0 \in X$ and $y_0 = f(x_0)$.
Let $\epsilon > 0$, by uniform continuity $\exists n$ such that $\forall n' \geq n_0, \forall x \in X$. $d(f_{n'}(x), f(x)) < \epsilon/3$.
Since f_n is continuous. Hence $\exists$ open $N \ni x_0$ such that $f_n(N) \subseteq B_{\epsilon/3}(f_n(x))$.

Hence $\forall x \in X d(f_n(x), f(x_0)) < \epsilon/3$, and $d(f_n(x), f(x)) < \epsilon/3$, and $d(f_n(x_0), f(x_0)) < \epsilon/3 \implies d(y_0, f(x)) < \epsilon$
Therefore, $f(x) \in B_\epsilon(y_0)$ and $f(N) \subseteq B_\epsilon(f(x_0))$ $\square$

Chapter 11

Gluing Constructions

Example 11.1.

- The unit circle $S^1 \subset \mathbb{R}^2$ is homeomorphic to the quotient $[0, 2\pi]/(0 \sim 2\pi)$ via $t \mapsto (\cos t, \sin t)$.
- The 2–sphere $S^2 = \{x \in \mathbb{R}^3 : \|x\|_2 = 1\}$ is homeomorphic to the quotient $D^2/\partial D^2$ obtained by collapsing the boundary of the closed disk D^2 to a point; equivalently, it is the space formed by gluing two copies of D^2 along their common boundary S^1.
- Identifying the left endpoints of three copies of $[0, 1]$—that is, forming $[0, 1] \vee [0, 1] \vee [0, 1]$ by imposing $0 \sim 0' \sim 0''$—yields a "three-armed star" graph (a 1-dimensional CW complex with one 0-cell and three 1-cells).
- Torus from a closed square: identify opposite edges with the same orientation.
- Klein bottle from a closed rectangle: identify one pair of opposite edges with the same orientation and the other pair with reversed orientation.

11.1 Two points of view

- Via a *quotient map*: let $q\colon [0, 2\pi] \to S^1$, $q(x) = (\cos x, \sin x)$. Declare $B \subseteq S^1$ open $\iff$ $q^{-1}(B)$ is open in $[0, 2\pi]$.
- Via an *equivalence relation*: on $X = [0, 2\pi]$ define $a \sim b$ if $a = b$ or $\{a, b\} = \{0, 2\pi\}$. Then $X/\sim = \{[x] : x \in X\}$ with $[0] = [2\pi]$. Declare $B \subseteq X/\sim$ open iff $\bigcup_{[x]\in B}[x]$ is open in X.

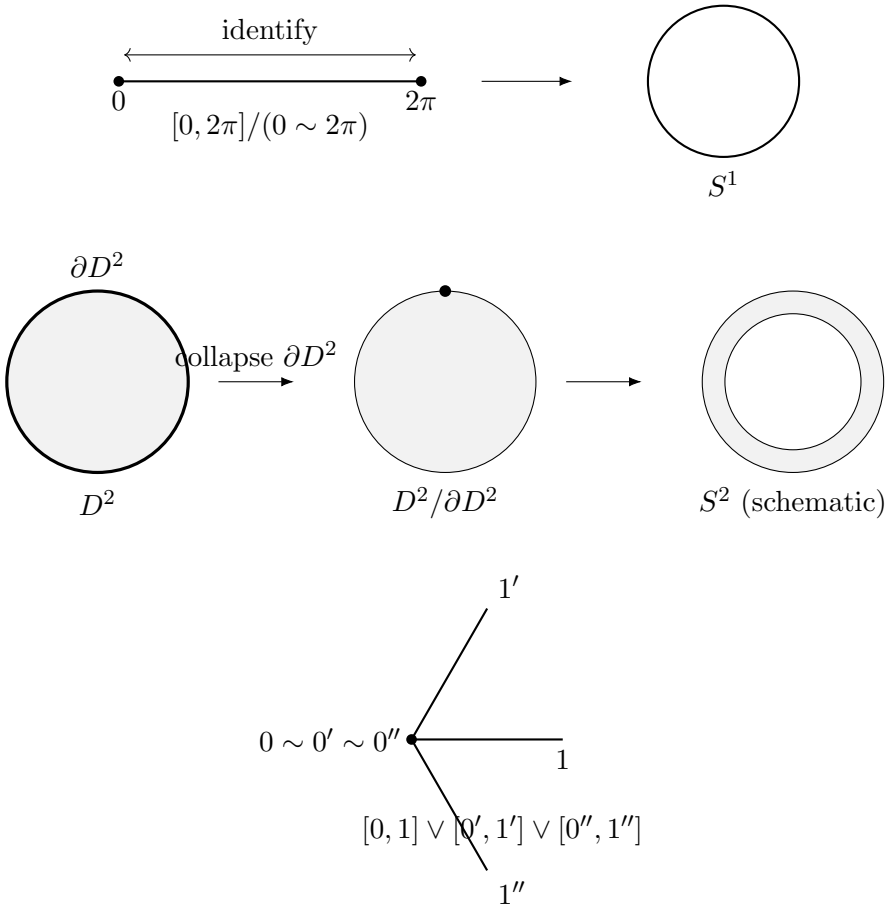


Figure 11.1: Topological identifications: (1) $S^1 \cong [0, 2\pi]/(0 \sim 2\pi)$; (2) $S^2 \cong D^2/\partial D^2$ (equivalently two disks glued along S^1); (3) wedge of three intervals.

Chapter 12

Strong and Weak Topologies

Definition 12.1 (Initial/Weak topology)**.** Let $f\colon X \to Y$ with Y a topological space. The *weakest* (initial) topology on X making f continuous is

$$\mathcal{T}_{\text{weak}} = \{\, f^{-1}(U) \subseteq X : U \subseteq Y \text{ open} \,\}.$$

This is indeed a topology on X (preimages preserve finite intersections and arbitrary unions).

Remark 12.2 (Final/Strong topology). Let $p\colon X \to Y$ with X a topological space. The *strongest* (final) topology on Y making p continuous is

$$\mathcal{T}_{\text{strong}} = \{\, B \subseteq Y : p^{-1}(B) \text{ is open in } X \,\}.$$

If p is surjective, this is the *quotient topology* and p is a *quotient map*.

Definition 12.3 (Quotient map)**.** A surjective map $f\colon X \to Y$ between topological spaces is a *quotient map* if for every $B \subseteq Y$,

$$B \text{ is open in } Y \iff f^{-1}(B) \text{ is open in } X.$$

Equivalently, Y carries the final topology with respect to f.

Definition 12.4 (Quotient space via an equivalence relation)**.** Let X be a topological space and $\sim$ an equivalence relation on X. Write $X/\!\sim = \{[x] : x \in X\}$ for the set of equivalence classes and let $q\colon X \to X/\!\sim$ be the canonical surjection $q(x) = [x]$. The *quotient topology* on $X/\!\sim$ is defined by

$$B \subseteq X/\!\sim \text{ is open} \iff q^{-1}(B) \text{ is open in } X.$$

(Equivalently, $q^{-1}(B) = \bigcup_{[x]\in B}[x]$ is open in X.)

Remark 12.5 (Quotients of Hausdorff spaces need not be Hausdorff). Even if X is Hausdorff, a quotient $X/\!\sim$ may fail to be Hausdorff. For example, let $X = [0,1] \times \{0,1\}$ (with the subspace topology from $\mathbb{R}^2$), and define $\sim$ by

$$(x,0) \sim (x,1) \text{ for all } x \in (0,1], \quad \text{while } (0,0) \not\sim (0,1).$$

Then the images of $(0,0)$ and $(0,1)$ in $X/\!\sim$ cannot be separated by disjoint open sets, so $X/\!\sim$ is not Hausdorff.

Claim. Every open $N \ni [(0,0)]$ and open $N \ni [(0,1)]$ intersect

Proof. Let $N \ni (0,0)$ be open in $X/\sim$, then $q^{-1}(N)$ is open in X. Since $(0,0) \in N$ we must have that $(0,0) \in q^{-1}(N)$ then $[0,\epsilon) \times \{0\} \subseteq q^{-1}(N)$ for some $\epsilon > 0$ $\forall x \in [0,\epsilon) \times \{0\}$ we must have $[x] \subseteq q^{-1}(N)$. In particular, $\forall a \in (0,\epsilon)$, since $(a,0) \in q^{-1}(N)$ we also have $(a,1) \in q^{-1}(N)$. Let $\epsilon > 0$ such that, $[0,\epsilon) \times \{0\} \subseteq q^{-1}(N)$ and $[0,\epsilon) \times \{1\} \subseteq q^{-1}(N')$. Then for every $0 < a < \epsilon$ we have $(a,0),(a,1) \in q^{-1}(N) \cap q^{-1}(N')$. Hence $[(0,0)] = [(a,1)] \in N \cap N'$. □

Proposition 12.6. *Let $q; X \to Y$ be a quotient map and $f : Y \to Z$. Then, f is continuous $\iff$ $f \circ g$ is continuous.*

Proof. ($\Longleftarrow$) $C \subseteq Z$ be open then $(f \circ g)^{-1}(C)$ is open in X which is the case $\iff$ $f^{-1}(C)$ is open in Y □

Question 12.7. Let $g : X \to Z$ be continuous. Does there exist a map $f : Y \to Z$ such that $h = f \circ g$ is unique

Chapter 13

Connected and Disconnected Spaces

Example 13.1. (intuition)

- $[0,1]$ is connected
- $[0,1) \cup [2,3]$ is disconnected
- $[0,1) \cup (1,2]$ is disconnected

Definition 13.2. A space X is disconnected if $\exists$ non-empty open A,B such that $A \cap B = \phi$ and $A \cup B = X$. We say X is connected if it is not disconnected.

Remark 13.3. A and B are clopen.

Remark 13.4. X is connected if it does not have any non-trivial clopen subset $\neq \emptyset, \neq X$

Example 13.5. (Connected spaces:)

- $\mathbb{R}$, $\mathbb{R}^d$, $[0,1]$, $(0,\infty)$, any interval
- $\mathbb{R}^2$ with lexicographic order
- Any ball in $\mathbb{R}^d$

Example 13.6. (Disconnected spaces:)

- $(0,1) \cup (1,2)$
- $\{0,1\}^{\mathbb{N}}$ (totally disconnected)
- $\mathbb{Q}$ (totally disconnected)
- $\mathbb{R} \setminus \mathbb{Q}$ (totally disconnected)
- Discrete space with more than one element is totally disconnected.
- evenly spaced topology of $\mathbb{Z}$ is totally disconnected

Definition 13.7. A space X is said to be totally disconnected if the only non-empty connected subsets of X are the singletons.

Proposition 13.8. *Any continuous image $f\colon X \to Y$ of a connected space is connected.*

Proof. Let $f\colon X \to Y$ be continuous and suppose, for contradiction, that $f(X)$ is disconnected. Then there exist disjoint nonempty open sets $U, V \subseteq f(X)$ (open in the subspace topology) such that

$$f(X) = U \cup V, \qquad U \cap V = \varnothing.$$

Because U and V are open in the subspace $f(X) \subseteq Y$, there exist open sets $U', V' \subseteq Y$ with

$$U = U' \cap f(X), \qquad V = V' \cap f(X).$$

Consider the preimages $A := f^{-1}(U)$ and $B := f^{-1}(V)$. Continuity of f implies

$$A = f^{-1}(U' \cap f(X)) = f^{-1}(U') \cap X, \qquad B = f^{-1}(V' \cap f(X)) = f^{-1}(V') \cap X,$$

so A and B are open in X. They are disjoint because U and V are, and they cover X since $f(X) = U \cup V$:

$$X = f^{-1}(f(X)) = f^{-1}(U \cup V) = A \cup B.$$

Moreover, A and B are nonempty because U and V are nonempty.

Thus X is the union of two disjoint, nonempty, open subsets, contradicting the connectedness of X. Hence $f(X)$ must be connected. □

Proposition 13.9. *Let X be any space, then the union of any collection of connected subsets of X that have a point in common is connected.*

Proposition 13.10. *Let X be any topological space. Then the union of any collection of connected subsets of X that have a point in common is connected.*

Proof. Let $\{A_i\}_{i \in I}$ be a collection of connected subsets of X such that there exists a point $x_0 \in \bigcap_{i \in I} A_i$. Let

$$A := \bigcup_{i \in I} A_i.$$

We will show that A is connected. Suppose, for contradiction, that A is disconnected. Then there exist nonempty disjoint open sets (in the subspace topology)

$$U, V \subseteq A, \qquad U \cap V = \varnothing, \quad U \cup V = A.$$

Since $A_i \subseteq A = U \cup V$, each A_i can be written as

$$A_i = (A_i \cap U) \cup (A_i \cap V),$$

where both $A_i \cap U$ and $A_i \cap V$ are open in the subspace topology of A_i.

Because A_i is connected, one of these intersections must be empty for each i. That is, for each i, either

$$A_i \subseteq U \quad \text{or} \quad A_i \subseteq V.$$

Now, since $x_0 \in A_i$ for all i, and U and V are disjoint, we have $x_0 \in U$ or $x_0 \in V$. Without loss of generality, suppose $x_0 \in U$. Then for every i, since $x_0 \in A_i \cap U \neq \varnothing$, we must have $A_i \subseteq U$.

Hence,

$$A = \bigcup_{i \in I} A_i \subseteq U,$$

which implies $V = \varnothing$, contradicting that both U and V are nonempty.

Therefore, A is connected. □

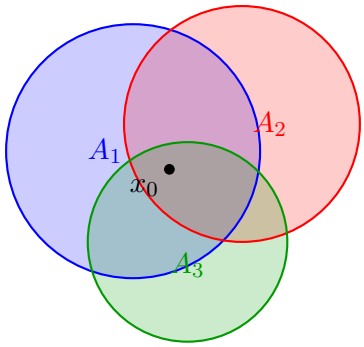


Figure 13.1: Several connected subsets $\{A_i\}$ sharing a common point x_0. Their union remains connected.

Definition 13.11. Two sets A and B are mutually separated if $\overline{A} \cap B = A \cap \overline{B} = \emptyset$

Proposition 13.12. *If $A, B \subseteq X$ are mutually separated and $A \cup B = X$ then A and B are clopen.*

Proposition 13.13. *If $A, B \subseteq X$ are mutually separated and $A \cup B = X$, then A and B are both open and closed (clopen) in X.*

Proof. By "mutually separated" we mean

$$\overline{A} \cap B = \varnothing \quad \text{and} \quad A \cap \overline{B} = \varnothing.$$

First, $A \cap B = \varnothing$, since $B \subseteq \overline{B}$ implies $A \cap B \subseteq A \cap \overline{B} = \varnothing$.

Because $A \cup B = X$ and $A \cap B = \varnothing$, we have $X \setminus B = A$ and $X \setminus A = B$.

From $\overline{A} \cap B = \varnothing$ we get $\overline{A} \subseteq X \setminus B = A$, hence $\overline{A} \subseteq A$, so A is closed. Symmetrically, $\overline{B} \cap A = \varnothing$ gives $\overline{B} \subseteq X \setminus A = B$, so B is closed.

For openness, since B is closed, its complement $X \setminus B = A$ is open; likewise, A closed implies $B = X \setminus A$ is open.

Thus A and B are both open and closed. □

Proposition 13.14. *$Y \subseteq X$ is disconnected $\iff$ there exist mutually separated sets $A, B \subseteq X$ such that $Y \subset A \cup B$, $A \cap Y \neq \varnothing$, and $B \cap Y \neq \varnothing$.*

Proof. ($\Rightarrow$) Assume Y is disconnected. Then $Y = C \cup D$ with C, D nonempty, disjoint, and mutually separated in Y. Choose open sets $U, V \subseteq X$ with $C = U \cap Y$ and $D = V \cap Y$. Since C and D are separated in Y, we have $C \subseteq Y \setminus \overline{D}^{Y}$ and $D \subseteq Y \setminus \overline{C}^{Y}$. Set

$$A := X \setminus \overline{V}, \qquad B := X \setminus \overline{U}.$$

Then A, B are open (hence mutually separated), $C \subseteq A \cap Y$ and $D \subseteq B \cap Y$, so $A \cap Y$ and $B \cap Y$ are nonempty, and

$$Y = C \cup D \subseteq (A \cap Y) \cup (B \cap Y) \subseteq (A \cup B) \cap Y,$$

whence $Y \subseteq A \cup B$.

($\Leftarrow$) Conversely, suppose $A, B \subseteq X$ are mutually separated with $Y \subseteq A \cup B$ and both $A \cap Y$ and $B \cap Y$ nonempty. Since A and B are mutually separated, the sets

$$U := Y \setminus \overline{B}^{Y} \quad \text{and} \quad V := Y \setminus \overline{A}^{Y}$$

are disjoint, open in Y, and satisfy $A \cap Y \subseteq U$, $B \cap Y \subseteq V$. Hence U and V are nonempty and

$$Y \subseteq (A \cup B) \cap Y \subseteq U \cup V.$$

Thus $Y = U \cup V$ is a separation, so Y is disconnected. □

Proposition 13.15. *Let $A, B \subseteq X$ be mutually separated and let $E \subset X$ be connected and $E \subseteq A \cup B$, then $E \subseteq A$ or $E \subseteq B$*

Proposition 13.16. *Let $A, B \subseteq X$ be mutually separated and let $E \subseteq X$ be connected with $E \subseteq A \cup B$. Then $E \subseteq A$ or $E \subseteq B$.*

Proof. Since A and B are mutually separated, we have

$$\overline{A} \cap B = \varnothing \quad \text{and} \quad A \cap \overline{B} = \varnothing,$$

where closures are taken in X. From $A \cap \overline{B} = \varnothing$ it follows that

$$A = (A \cup B) \setminus \overline{B},$$

so A is open in the subspace $A \cup B$; similarly, B is open in $A \cup B$. Hence the sets

$$U := E \cap A \quad \text{and} \quad V := E \cap B$$

are disjoint open subsets of E with $E = U \cup V$.

If both U and V were nonempty, this would separate E, contradicting its connectedness. Therefore one of U, V is empty. If $V = \varnothing$ then $E \subseteq A$; if $U = \varnothing$ then $E \subseteq B$. □

Proposition 13.17. *If $E \subseteq X$ is connected and $E \subseteq F \subseteq \overline{E}$, then F is connected.*

Proof. Suppose, for a contradiction, that F is disconnected. Then there exist disjoint, nonempty sets U, V open in the subspace F with $F = U \cup V$.

Since $E \subseteq F$ and E is connected, we must have $E \subseteq U$ or $E \subseteq V$. Without loss of generality, assume $E \subseteq U$. Then $E \cap V = \varnothing$.

Because V is open in F, there exists an open set $O \subseteq X$ with $V = O \cap F$. If $V \neq \varnothing$, pick $y \in V \subseteq F \subseteq \overline{E}$. As $y \in \overline{E}$, every neighborhood of y in X meets E, hence $O \cap E \neq \varnothing$. But

$$O \cap E = (O \cap F) \cap E = V \cap E = \varnothing,$$

a contradiction. Therefore $V = \varnothing$, and similarly $U \neq \varnothing$ cannot fail.

Hence no such separation exists and F is connected. □

Theorem 13.18. *Let $\{X_i\}_{i \in I}$ be a family of topological spaces. Then*

$$\Big(\forall i \in I,\ X_i \text{ connected}\Big) \iff \prod_{i \in I} X_i \text{ is connected}$$

with the product topology.

Proof. **Special case:** $|I| = 2$. Let X, Y be connected. Suppose $X \times Y$ is not connected. Then there exist non-empty, disjoint open subsets $A, B \subseteq X \times Y$ such that $A \cup B = X \times Y$.

Claim. Either there exists $x_0 \in X$ such that $\{x_0\} \times Y$ intersects both A and B, or there exists $y_0 \in Y$ such that $X \times \{y_0\}$ intersects both A and B.

WLOG, assume the second is true.

Define

$$A_{y_0} = \{x \in X : (x, y_0) \in A\} \neq \emptyset, \qquad B_{y_0} = \{x \in X : (x, y_0) \in B\} \neq \emptyset.$$

Claim. A_{y_0}, B_{y_0} are open in X. Moreover, $A_{y_0} \cap B_{y_0} = \emptyset$ and $A_{y_0} \cup B_{y_0} = X$. Hence X is disconnected.

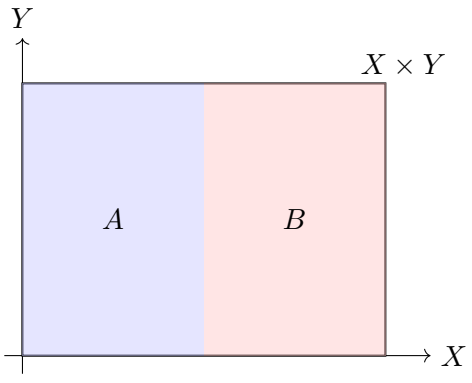
Proof of Claim 2. Let $x \in A_{y_0}$. Then $(x, y_0) \in A$. Since A is open, there exist open sets $V \subseteq X$ and $W \subseteq Y$ such that

$$(x, y_0) \in V \times W \subseteq A.$$

Therefore $x \in V \subseteq A_{y_0}$, so A_{y_0} is open. Similarly, B_{y_0} is open. □

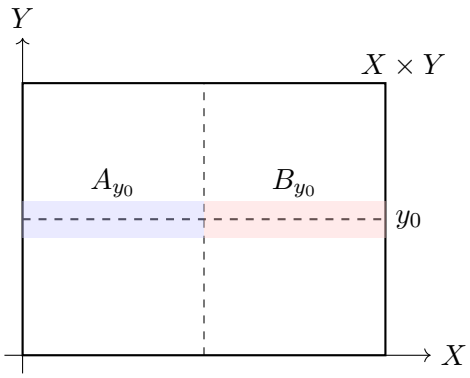
Proof of Claim 1. Suppose the first alternative is not true. Let $(x_1, y_1) \in A$ and $(x_2, y_2) \in B$. Consider (x_1, y_2). Since $(x_1, y_1) \in A$, by the product topology there exists an open rectangle around (x_1, y_1) contained in A, hence $(x_1, y_2) \in A$. Thus $X \times \{y_2\}$ intersects both A and B. □

□



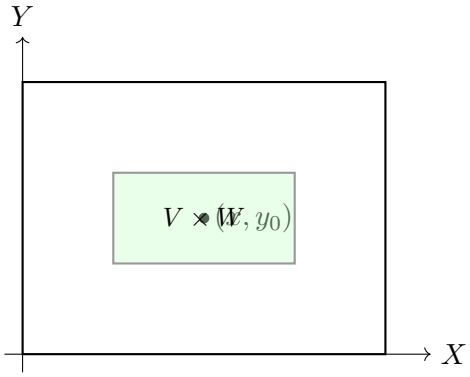
A, B are open, disjoint, and cover $X \times Y$

Figure 13.2: Hypothetical disconnection of $X \times Y$ into disjoint open sets A and B.



Cross-section $X \times \{y_0\}$ intersects both A and B.

Figure 13.3: Horizontal section showing induced open sets $A_{y_0}, B_{y_0} \subseteq X$.



Open rectangle $V \times W \subseteq A$ witnessing openness in the product topology.

Figure 13.4: Neighborhood structure around (x, y_0) in the product topology.

Theorem 13.19. *General Case: Case1:* $\prod_{i\in I} X_i = \emptyset$ *Case 2:* $\prod_{i\in I} X_i \neq \emptyset$. *Let* $a \in \prod_{i\in I} X_i$ *For every finite* $J \subseteq I$, *let* $\tilde{X}_J = \{x \in X : \forall i \notin J, x_i = a\}$ *Note* $\tilde{X}_J \cong \prod_{i\in J} X_i$ *Connected. Since* $a \in \bigcap_{J\subseteq I \text{(finite)}} \tilde{X}_J$, *by (2),* $\bigcup_{J\subseteq I} \tilde{X}_J$ *is connected.*

Proposition 13.20. *The union* $\bigcup_{J\subseteq I \text{ finite}} X_J$ *is dense in* X.

Example 13.21. Let $X = \{0,1\}^{\mathbb{N}}$.

By (6), the closure $\overline{\bigcup_{J\subseteq I \text{ finite}} X_J}$ is connected. □

Proof of Proposition. Let B be a basis element by the product topology on Y. $B = \bigcap_{k\in J} \pi_k^{-1}(U_k)$ for some finite $J \subseteq I$ and open sets $U_k \subseteq X_k$. $B \cap X_J \neq \emptyset$. Let $b \in B$ define x with:

$$x_i = \begin{cases} b_i, & i \in J, \\ a_i, & i \notin J \end{cases}$$

then $x \in B$ and $x \in \tilde{X}_J$ □

13.1 Connected Subcomponents

Let X be a topological Space. We write $x \sim y$ if $\exists$ connected $C \subseteq X$ such that $x, y, z \in C.x \sim y$ and $y \sim z \implies x \sim z$

Definition 13.22. The equivalence classes of $\sim$ are called the *connected components* of X

Proposition 13.23. *Each connected component of X is connected.*

Proof. Let $C \subseteq X$ be a component. Let $A, B \in X$ be mutually separated with $C \subseteq A \cup B$ such that C intersects both A and B. Pick $x \in A \cap C$ and $y \in B \cap X$. Since $x \sim y$, $\exists$ connected $E \subseteq X$ such that $x, y \in E$. Observe that $E \subseteq C$. By (5), either $E \subseteq A$ or $E \subseteq B$ □

Remark 13.24. For $x \in X$, the connected component $C(x)$ of x is the largest connected subset of X that contains x.

Example 13.25. (1) $(0,1) \cup (1,2)$ are $(0,1), (1,2)$

(2) the connected components of $\mathbb{Q}$ are the singletons.

Proposition 13.26. *Each connected component of X is closed in X*

Proof. Let $C \subseteq X$ be a connected component. by (6), $\overline{C}$ is also connected; hence, $\overline{C} = C$, which means C is closed. □

Proposition 13.27. *A set $I \subseteq \mathbb{R}$ is connected $\iff$ it is an interval*

Proof. ($\implies$) If I is not an interval means i can find $a, b \in I$ and $c \notin I$ such that $a < c < b$ then $(-\infty, c) \cap I$ and $(c, \infty) \cap I$ form a nontrivial open partition of I.
($\impliedby$) Let I be an interval, suppose $\exists A, b \subseteq I$ nonempty, disjoint open in I with $A \cup B = I$. Let $a \in A$ and $b \in B$. Let $c = sup\{x \in A : x < b\}$ Now is $c \in A$ or $c \in B$. My claim is $c \in \partial A$ contradiction. Let $\epsilon > 0$ then $\exists x \in (c - \epsilon, c)$ such that $x \in A$, $\forall y \in [c, c + \epsilon) \cap I, y \in B$ □

Theorem 13.28 (intermediate value theorem)**.** *Let X be a connected space and $f \colon X \to \mathbb{R}$ continuous. Let $a, b \in X$ be distinct then c takes all the values between $f(a)$ and $f(b)$*

Definition 13.29. A space X is *pathwise connected* if $\forall a, b \in X, \exists\ \gamma : [0,1] \to X$ continuous such that $\gamma(0) = a$ and $\gamma(1) = b$

Proposition 13.30. *Every path connected space is connected.*

Example 13.31 (connected but not path-connected)**.** $X = \{(x, sin(1/x) : x \in (0, \infty)\} \subseteq \mathbb{R}^2$ $\overline{X} = X \cup \{(0, y) : y \in [-1, 1]\}$ $\overline{X}$ is connected but it is not path connected

Proof. Suppose $\exists$ continuous $\gamma : [0,1] \to X$ such that $\gamma(0) = (1, sin1)$ and $\gamma(1) = (0.0)$. Let $x(t)$ and $y(t)$ be the coordinates of $\gamma(t)$. Note that $x : [0,1] \to [0, \infty)$ and $y : [0,1] \to [-1, +1]$ are continuous. Set $a = inf\{t \geq 0 : x(t) = 0\} \neq \emptyset$ since x is continuous, $x(a) = 0$ as $t \to a^-$ we must have $x(t) \to 0^+$ and $y(t) \to y(a)$.
Claim. $\forall b \in [-1, 1], \exists t_n \to a^-$ such that $y(t_n) = b$.

Let $\theta \in [-\pi/2, \pi/2]$ such that $sin(\theta) = b$ and $sin(2\pi m + \theta) = b, \forall m \in \mathbb{Z}$ Note: for $t < a, y(t) = sin(\frac{1}{x(t)})$. So for each n, consider $(a - \frac{a}{2^n}, a)$. Now lets pick m_n such that $0 < \frac{1}{2\pi m_n + \theta} < x(a - \frac{a}{2^n}$ by IVT, $\exists t_n \in (a - \frac{a}{2^n}, a)$ such that $x(t_n) = \frac{1}{2\pi m_n + \theta}$. So $y(t_n) = sin(\frac{1}{x(t_n)} = sin(2\pi m_n + \theta) = b$ □

Chapter 14

Compactness

Definition 14.1. Let X be a topological space and $E \subseteq X$. A cover of E in X is a family $U \subseteq 2^X$ such that $\bigcup_{A \in U} A \supseteq E$.

- if every $A \in U$ is open we call U an open cover.
- A sub-cover of U is a subfamily $U' \subseteq U$ which is a again a cover.

Definition 14.2. A space X is *compact* if *every open cover* of X has a *finite subcover*.

Example 14.3.

- $\mathbb{R}$ is not compact. Take $U = \{(-n, n) : n > 0\}$
- $[0, 1]$ is compact.

  Proof. Let U be an open cover of $[0, 1]$. Define

  $$S = \{x \in [0, 1] : [0, x] \text{ is covered by a finite subfamily of } U\}.$$

  Note that $0 \in S$ and, if $x \in S$ and $0 \leq y \leq x$, then $y \in S$. Take $e = \sup S$. Then $[0, e) \subseteq S$ and $e \in [0, 1]$.

  Claim. $e \in S$.

  Proof. Let $A^* \in U$ such that $A^* \in e$. Since A^* is open $\exists \epsilon > 0$ such that $(e-\epsilon, e+\epsilon) \subseteq A^*$. Since $e - \epsilon \in S_1, \exists A_1, A_2, \ldots A_n \in U$ such that $\bigcup_{i=1}^{n} A_i \supseteq [0, e-\epsilon]$ then, $A_1, A_2, \ldots A_n, A^*$covers$[0, e]$. Note: The same argument shows that $\forall e' \in (e, e + \epsilon) \cap [0, 1]$. We have $e' \in S$. since $e = supS$ we find that $(e, e + \epsilon) \cap [0, 1] = \emptyset$ which means $e = 1$ □

  □

- $\{0, 1\}^{\mathbb{N}}$ compact by Tychonov
- $[0, 1]^{\mathbb{N}}$ compact by Tychonov
- $[0, 1]^{\mathbb{R}}$ compact by Tychonov
- S^1, S^n compact
- Cantor space is compact by heine-borel

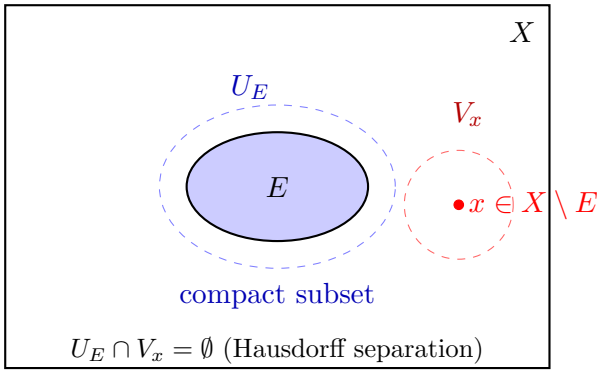
Proposition 14.4. *X compact, $E \subseteq X$ closed $\implies$ E compact.*

Proof. Let U be an open cover of E in X. Then, $U' = U \cup X \setminus E$ is an open cover of X. Since X is compact, U' has a finite subcover. That is, there exists $A_1, A_2, \ldots, A_n \in U$ such that $A_1 \cup A_2 \cup \cdots \cup A_n \cup (X \setminus E) \supset X$ it follows that $A_1 \cup A_2 \cup \cdots \cup A_n \supseteq E$. Thus U has a finite subcover of E. □

Remark 14.5. The converse is not true in general. $\{0, 1\}$ with trivial topology $\{0\}$ is compact but not closed.

Proposition 14.6. *Suppose X is Hausdorff and $E \subseteq E$ is compact $\implies$ E is closed in X*

Proof. Let $x \in X \setminus E$, we show that x has an open neighborhoodthat does not intersect E. Since X is Hausdorff, $\forall y \in E$, $\exists V_y \ni x$ open and $W_y \ni y$ open such that $V_y \cap W_y = \emptyset$ For each $y \in E$. fix such V_y and W_y Note: $\{W_y : y \in E\}$ is an open cover of E. Since E is compact, $\exists$ finite $I \subseteq E$ such that $\bigcup_{y \in I} W_y \supset E$ Let $N = \bigcap_{y \in I} V_y \ni x$ then N is open and $N \cap W_y = \emptyset$ for $y \in I$, hence $\emptyset = N \cap (\bigcup_{y \in I} W_y) \supset N \cap E$ □

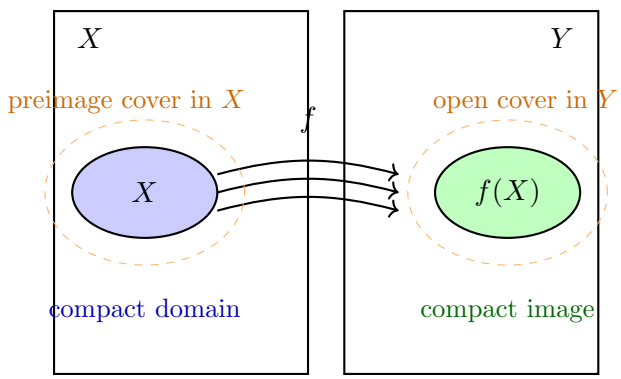


Proposition 14.7. *Let X be a Hausdorff. Let $E, F \subseteq X$ be disjoint and compact. Then $\exists$ open $N \supseteq E$ and $\exists$ open $M \supseteq F$ such that $N \cap M = \emptyset$*

Proof. Apply the trick from last proof twice. □

Proposition 14.8. *X compact and $f \colon X \to Y$ continuous $\implies$ $f(X)$ is compact*

Proof. Let U be an open cover of $f(x)$, then $W = \{f^{-1}(B) : B \in U\}$ is an open cover of X. Since X is compact, $\exists B_1, B_2, \ldots B_n \in U$ such that $\bigcup_{i=1}^{n} f^{-1}(B_i) \supseteq X$, then for such $B_1, B_2, \ldots B_n$ we have $\bigcup_{i=1}^{n} B_i \supseteq f(X)$ □



Theorem 14.9. *(Tychonoff) The product of any family of compact spaces is compact.*

$$X_i \text{ compact} \iff \prod_{i\in I} X_i \text{ compact}$$

Example 14.10.

- Any closed hypercube, $[0,1]^n \subseteq \mathbb{R}^n$
- $\{0,1\}^{\mathbb{N}}$
- $S^1 \times S_1$ (Torus)
- $[0,1]^{\mathbb{R}}$

Proof. X, Y compact $\implies$ $X \times Y$ compact. Let C be an open cover of $X \times Y$. Let $y \in Y$, and define $X_y = X \times \{y\}$. Note $X_y \cong X$, hence X_y is compact. Also note that C covers X_y, so $\exists$ finite $C_y^\circ \subset C$ that covers X_y. Let $V_y = \bigcup_{A\in C_y^\circ} A$; note that V_y is open.

Claim. $\exists$ open $B_y \ni y$ such that $X_y \subseteq X \times B_y \subseteq V_y$.

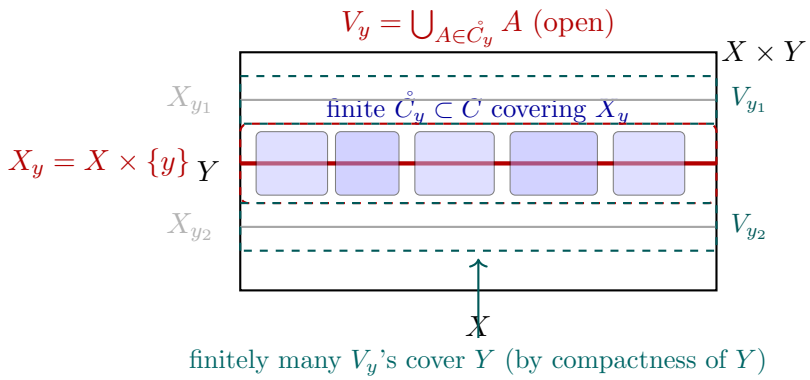
Assuming the claim, let $E = \{B_y : y \in Y\}$. This is an open cover of Y, and since Y is compact, $\exists$ finite subcover $Y_0 \subseteq Y$ such that $E_0 = \{B_y : y \in Y_0\}$ covers Y. Therefore,

$$X \times Y \subseteq \bigcup_{y\in Y_0} X \times B_y \subseteq \bigcup_{y\in Y_0} V_y \subseteq \bigcup_{y\in Y_0} \bigcup_{A\in C_y^\circ} A.$$

Hence $C^\circ = \bigcup_{y\in Y_0} C_y^\circ$ is a finite subcover of $X \times Y$.

Proof of claim. Since V_y is open, $\forall x \in X$, $(x,y) \in V_y$ $\exists$ open $I_{x,y} \subseteq X \times Y$ and open $J_{x,y} \subseteq Y$ such that $(x,y) \in I_{x,y} \times J_{x,y} \subseteq V_y$. Now $D_y = \{I_{x,y} \times J_{x,y} : x \in X\}$ is an open cover of X_y. Hence, $\exists$ finite $D_y^\circ \subseteq D_y$ that covers X_y. Let $B_y = \bigcap\{J_{x,y} : I_{x,y} \times J_{x,y} \in D_y\}$ then, B_y is open and $X \times B_y \subseteq V_y$. $\forall x \in X, \exists x'$ such that $I_{x',y} \times J_{x',y} \in D_y^\circ$ and $x \in I_{x',y}$, hence $\{x\} \times B_y \in I_{x',y} \times J_{x',y} \subseteq V_y$ □

□



14.1 Compactness in terms of basis and subbasis

Proposition 14.11. *Let X be a topological space with basis $\mathcal{B}$, and $E \subseteq X$. Then E is compact $\iff$ every cover of E by elements of $\mathcal{B}$ has a finite subcover.*

Proof. ($\Longrightarrow$) Trivial. ($\Longleftarrow$) Let $U = \{A_i : i \in I\}$ be an open cover of E. Given $i \in I$ and $x \in A_i$, let $B_{i,x} \subseteq A_i$. Now, $\mathcal{C} = \{B_{i,x} : i \in I, x \in A_i\}$ is a $\mathcal{B}$-cover of E. By assumption, $\exists$ finite $\mathcal{C}^\circ \subseteq \mathcal{C}$ that covers E. For each $B_{i,x} \in \mathcal{C}^\circ$, pick the corresponding $A_i \in U$. The resulting finite subfamily of U covers E. □

Theorem 14.12 (Alexander subbasis theorem)**.** *Let X be a topological space with subbasis $\mathcal{S}$, and $E \subseteq X$. Then E is compact $\iff$ every cover of E by elements of $\mathcal{S}$ has a finite subcover.*

Example 14.13.

- $[0, 1]$ is compact,take a subbasis $\mathcal{S} = \{(-\infty, b) : b \in \mathbb{R}\} \cup \{(a, \infty) : a \in \mathbb{R}\}$
- proof of Tychonoff theorem using subbasis theorem.

  Proof. Suppose $X_i (i \in I)$ are compact. Let $X = \prod_{i \in I} X_i$. Recall that the subbasis of the product topology is given by

  $$\mathcal{S} = \{\pi_i^{-1}(U) : i \in I, U \subseteq X_i \text{ open}\}.$$

  Let $C = \{\pi_i^{-1}(U) : i \in I, U \subseteq U_i$ family of open subsets of $X_i\}$. Consider 2 cases:

  - Case 1: There exists $i \in I$ such that U_i covers X_i. Since X_i is compact, we can find finite $U_i^0 \subseteq U_i$ that covers X_i. Then $C^0 = \{\pi_i^{-1}(U) : U \in U_i^0\} \subseteq C$ is a finite subcover of X.
  - Case 2: $\forall i \in I, \exists x_i \in X_i$but $x_i \notin U$ for any $U \in U_i$ fix such an x_i for each i. Let $X = (x_i)_{i \in I}$ is not covered by C, because if $c \in \pi^{-1}(U)$ for some $U \in U_i$, then $x_i \in U$, which is not possible, hence contradiction.

  Therefore C has a finite subcover of X. By Alexander subbasis theorem, X is compact. □

14.2 Compactness in terms of sequences

- For $E \in \mathbb{R}^n$, the following are equivalent:

  (1) every open cover of E has a finite subcover (compactness).

  (2) every sequence in E has a convergent subsequence with its limit in E.(sequential compactness).

  (3) E is closed and bounded. (Heine-Borel)

  For general topological spaces, (1) and (2), but not equivalent.

  Example 14.14.
  - $\{0, 1\}^{\mathbb{R}}$ is compact but not sequentially compact.
  - $\{x \in \{0, 1\}^{\mathbb{R}} : x_i = 0$ for all but countably many $i\}$ is sequentially compact but not compact. Here it is enough to show that it is not closed.

Theorem 14.15. *Let X be a metrizable space. Then X is compact iff X is sequentially compact.*

Proof. Let d be a metric that for the topology of X ($\Longrightarrow$) Suppose X is not sequentially compact. Then *exists* a sequence $x_1, x_2, \cdots \in X$ with no convergent subsequence.

Claim. $\forall x \in X, \exists \epsilon > 0$ such that $B_\epsilon(x)$ contains at most finitely many x_i's.For each $x \in X$, let $\epsilon_x > 0$ such that $B_{\epsilon_x}(x)$ have the above property.

Proof. Suppose not, let x be a point for which the claim doesn't hold. $\epsilon = 1$, $I_1 = \{n \in \mathbb{N} : x_n \in B_1(x)\}$ infinite $\epsilon = 1/2$, $I_2 = \{n \in I_1 : x_n \in B_{1/2}(x)\}$ infinite $\dots$ $\epsilon = 1/k$, $I_k = \{n \in I_{k-1} : x_n \in B_{1/k}(x)\}$ infinite. Let $n_1 = minI_2, n_2 = minI_2 \setminus \{n_1\}, \dots n_k = minI_k \setminus \{n_1, n_2, \dots n_{k-1}\}$. Let $I = \{n_1, n_2, \dots, n_k\}$, since $\forall k I \setminus I_k$ is finite,$\exists I_0$ such that $\forall i \geq i_0, n_i \in I_k$, so $x_{n_i} \in B_{1/k}(x)$. Hence $x_{n_i} \to x$ contradiction. □

Claim. $C = \{B_{\epsilon_x} : x \in X\}$ is an open cover of X without a finite subcover.

Proof. If $C_0 \subseteq C$ is finite and covers X then by the pidgeon hole principle, some $B_{\epsilon_x} \in C_0$ contains infinitely many x_i's contradiction. □

($\Longleftarrow$) Suppose X is sequentially compact. Let U be an open cover of X.

Claim. $\exists \epsilon > 0$ such that $\forall x \in X$, then $B_\epsilon(x) \subseteq A$ for some $A \in U$

Claim. $\forall \epsilon > 0$, X can be covered by finitely many ϵ-balls.

Assuming the claims, let $\epsilon > 0$ be as in the first claim. By the second claim, $\exists x_1, x_2, \dots, x_n \in X$ such that $X \subseteq \bigcup_{i=1}^{n} B_\epsilon(x_i)$. By the first claim, $\forall i$, $B_\epsilon(x_i) \subseteq A_i$ for some $A_i \in U$. Therefore, $A_1, A_2, \dots, A_n$ is a finite subcover of U.

Proof of Second claim. Suppose the claim does not hold for some $\epsilon > 0$,

$$\begin{aligned}
&\text{Pick } x_1 \in X,\\
&\text{Pick } x_2 \in X \setminus B_\epsilon(x_1),\\
&\text{Pick } x_3 \in X \setminus \big(B_\epsilon(x_1) \cup B_\epsilon(x_2)\big),\\
&\quad\vdots\\
&\text{Pick } x_n \in X \setminus \big(B_\epsilon(x_1) \cup B_\epsilon(x_2) \cup \cdots \cup B_\epsilon(x_{n-1})\big).
\end{aligned}$$

By assumption this can be done for every n, but for i, j distinct, $d(x_i, x_j) \geq \epsilon$, hence no subsequence of $(x_n)_{n\geq 0}$ is cauchy. Therefore X is not sequentially compact contradiction. □

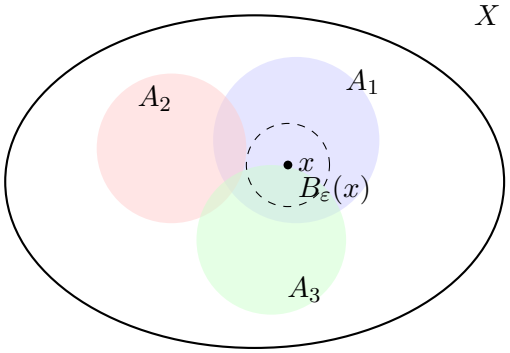
Proof of First claim. Suppose not, $\forall n \in \mathbb{N}, \exists x_n \in X$ such that $B_{1/n}(x_n) \not\subseteq A$ for any $A \in U$. Since X is sequentially compact, $\exists$ a subsequence $(x_{n_k})_{k\geq 0}$ that converges to some $x \in X$. Since U covers X, $\exists A^* \in U$ such that $x \in A^*$. Since A^* is open, $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq A^*$. Pick K such that $1/n_K < \epsilon$ and $d(x_{n_K}, x) < \epsilon/2$, then for any $y \in B_{1/n_K}(x_{n_K})$, we have

$$d(y, x) \leq d(y, x_{n_K}) + d(x_{n_K}, x) < 1/n_K + \epsilon/2 < \epsilon.$$

Hence, $B_{1/n_K}(x_{n_K}) \subseteq B_\epsilon(x) \subseteq A^*$ contradiction.

□

□



Theorem 14.16. *Let X be a metric space and $E \subseteq X$, then E is compact $\iff$ E is complete and totally bounded.*

Proof. — ask a toddler on the street □

Theorem 14.17 (Lebesgue number covering lemma)**.** *Let X be a compact metric space and U be an open cover of X. There exists $\delta > 0$ such that if $C \subset X$ is a set with diameter less than δ, then $C \subseteq A$ for some $A \in U$.*

Proof. — ask a different toddler on the street. □

Remark 14.18. remember that $diam(C) = sup\{d(x, y) : x, y \in C\}$. and every open cover of a compact space has a Lebesgue number.

14.3 Compactness in terms of closed sets

Definition 14.19. A family $\mathcal{C}$ of subsets of X has the finite intersection property, if every finite subfamily of $\mathcal{C}$ has non-empty intersection. $C_1, C_2, \ldots, C_n \in \mathcal{C} \implies \bigcap_{i=1}^{n} C_i \neq \emptyset$

Proposition 14.20. *Let X be a topological space, then X is compact $\iff$ every family $\mathcal{C} \subseteq 2^x$ of closed sets with the FIP has a nonempty intersection.*

Example 14.21. Let X be a collection of boolean variables and S be a collection of propositions. If every finite subfamily of S is satisfiable then S is satisfiable

Corollary 14.22 (Nested interval theorem)**.** *Let X be a compact topological space and $C_1 \supseteq C_2 \supseteq C_3 \supseteq \ldots$ be a sequence of non-empty closed subsets of X. Then, $\bigcap_{n=1}^{\infty} C_n \neq \emptyset$.*

Proof. Note that $\{C_n : n \in \mathbb{N}\}$ has the FIP, hence by the previous proposition, $\bigcap_{n=1}^{\infty} C_n \neq \emptyset$. □

Example 14.23 (Counter example)**.**

- $X = \mathbb{R}$ and $C_n = (n, \infty)$ then $\bigcap_{n=1}^{\infty} C_n = \emptyset$
- $X = \mathbb{Q}$ and $C_n = [\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}] \cap \mathbb{Q}$ then $\bigcap_{n=1}^{\infty} C_n = \emptyset$

Theorem 14.24 (Extreme value theorem)**.** *Let X be a compact topological space and $f \colon X \to \mathbb{R}$ be continous. Then, f attains a minimum and a maximum.*

Proof. — Math 210 □

Remark 14.25. if $f \colon X \to Y$ is bijective and continuous, with X compact and Y Hausdorff, then f is a homeomorphism.

Proposition 14.26. *$f \colon X \to Y$ is a bijection, X compact, Y Hausdorff $\implies$ f is a homeomorphism.*

Proof. Want to show that f^{-1} is continuous. Let $E \subseteq X$ be closed, we show that $(f^{-1})^{-1}(E)) = f(E)$ is closed in Y. Note that $E \subseteq X \implies E$ Compact $\implies f(E)$ Compact $\implies f(E)$ Closed $\implies$ $f(E)$ Closed. □

Example 14.27. Cantor space: defined by $Y = \bigcap_{n=0}^{\infty} y_N$ where $y_0 = [0, 1]$ and y_n is obtained from y_{n-1} by removing the open middle third of each interval in y_{n-1}. Then Y is compact as a closed subset of $[0, 1]$. Define $\phi \colon \{0, 1\}^{\mathbb{N}} \to Y$ by $\phi(x_1, x_2, x_3, \ldots) = \sum_{n=1}^{\infty} (\frac{2}{3})^n x_n$.

Remark 14.28. Recall that if X and Y are metric and $f \colon X \to Y$ is uniformly continuous if $\forall \epsilon > 0, \exists \delta > 0$ such that$d(x, y) < \delta \implies d(f(x), f(y)) < \epsilon$

Proposition 14.29. *X and Y metric, $f\colon X \to Y$ continuous, X compact $\implies$ f is uniformly continuous.*

Proof. Let $\epsilon > 0$. Since f is continous, $\forall z \in X \exists \delta_z > 0$ such that $f(B_{\delta_z}(z)) \subseteq B_{\epsilon/2}(f(z))$. Note that $U = \{B_{\delta_z}(z) : z \in X\}$ is an open cover of X. Let δ be a Lebesgue number for U. Let $x, y \in X$ such that $d(x,y) < \delta$ then $\exists z \in X$ such that $B_{\delta_z}(z) \supseteq \{x, y\}$ so $d(x,z) < \delta_z$ and $d(y,z) < \delta_z$ so $d(f(x), f(z)) < \epsilon/2$ and $d(f(y), f(z)) < \epsilon/2$ hence by triangle inequality $d(f(x), f(y)) < \epsilon$. Since ϵ was arbitrary, f is uniformly continuous. □

14.4 Local Compactness

Definition 14.30. X is *locally compact* if every point $x \in X$ has a compact neighborhood.

Remark 14.31. Recall that if $\mathcal{B} \subseteq 2^X$ is a neighborhood basis for a point x if for every open $N \ni x$ $\exists B \in \mathcal{B}$ such that $x \in B \subseteq N$.

Proposition 14.32. *Let X be a Hausdorff. X is locally compact $\iff$ every $x \in X$ has a neighborhood basis consisting of compact sets.*

Proof.

Lemma 14.33. *X Hausdorff, Let $A, B \subseteq X$ be disjoint and compact. Then, $\exists$ open $N \supseteq A, \overline{N} \cap B \neq \emptyset$*

Proof. — Hw. 6 □

($\implies$) Let $x \in X$ by assumption $\exists$ compact set C such that $x \in C^\circ$. Let A be an open neighborhood of x. WLOG we can assume $A \subseteq C$ (if not replace it with $A \cap C^\circ$). Note that $C \setminus A$ is compact so by the lemma, $\exists$ open $N \ni x$ such that $N \cap (C \setminus A) = \emptyset$. Let $D = N \cap C^\circ$ which is open.

Claim. $x \in D$, D is open, $\overline{D} \subseteq A$ and $\overline{D}$ is compact.

Proof. $x \in D$ since $x \in N$ and $x \in C^\circ$. D is open as intersection of open sets. $\overline{D} \subseteq \overline{N} = N$ since N is open. $\overline{D} \subseteq \overline{C^\circ} \subseteq C$ since C is closed, hence $\overline{D}$ is compact as a closed subset of a compact set. $\overline{D} \subseteq A$ since if $y \in \overline{D}$ then $y \in N$ and $y \in C$, hence $y \notin C \setminus A$ so $y \in A$. but $\overline{D} = \overline{N \cap C^\circ} \subseteq A$ □

□

Proposition 14.34. *Let X be locally compact Hausdorff.*

- *$A \subseteq X$ closed $\implies$ A is locally compact.*
- *$U \subseteq X$ open $\implies$ U is locally compact.*
- *$A, B \subseteq X$ locally compact $\implies$ $A \cap B$ is locally compact.*

Proof.

- Let $x \in A$, since X is locally compact, $\exists$ compact neighborhood C of x in X. Note that $C \cap A$ is a compact neighborhood of x in A.
- Let $x \in U$, since X is locally compact, $\exists$ compact neighborhood C of x in X. Note that $C \cap U$ is a compact neighborhood of x in U.

- Let $x \in A \cap B$, since A is locally compact, $\exists$ compact neighborhood C_A of x in A. Since B is locally compact, $\exists$ compact neighborhood C_B of x in B. Note that $C_A \cap C_B$ is a compact neighborhood of x in $A \cap B$.

□

Remark 14.35. The continuous image of a locally compact space need not be locally compact.

Example 14.36. Consider $\mathbb{R}^2$ with the map $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) = xy$. Note that $\mathbb{R}^2$ is locally compact but $f(\mathbb{R}^2) = \mathbb{R}$ is not locally compact.

Definition 14.37. A map $f\colon X \to Y$ is an *open map* if it maps open sets to open sets.

Proposition 14.38. *If X is locally compact and $f\colon X \to Y$ is continuous and open, then $f(X)$ is locally compact.*

Proof. Let $y \in f(x)$. Let $x \in X$ be such that $f(x) = y$. Since X is locally compact, Let $C \ni x$ be a compact neighborhood of x.

- $f(C)$ is compact
- $f(C^\circ)$ is open
- $y \in f(C^\circ) \subseteq f(C)$
- Hence, $f(C)$ is a compact neighborhood of y in $f(X)$.

□

14.5 Compactification

Example 14.39. .

- $\mathbb{R}$ not compact but $\mathbb{R}^* = \mathbb{R} \cup \{\infty\}$ is compact. since $\mathbb{R} \cong S^1 \setminus \{p\}$ via stereographic projection.
- $\mathbb{R}^n$ not compact but $\mathbb{R}^n \cong S^n \setminus \{p\}$ is compact.
- $\mathbb{R}$ not compact but $\mathbb{R} \cup \{-\infty, \infty\} \cong [0,1]$ is compact.

Definition 14.40. A *compactification* of a topological space X is and *embedding* $\phi\colon \tilde{X}$ into a compact Hausdorff space such that $\phi(X)$ is dense in $\tilde{X}$.

14.5.1 One point Compactification

X Hausdorff, locally compact, non-compact. Define $X^* = X \cup \{\infty\}$. The embedding $\phi\colon X \to X^*$ where $\phi^*(x) = x$. The topology on X^* is defined as $\mathcal{T}^* = \mathcal{T} \cup \{X^* \setminus C : C \subseteq X \text{ compact}\}$
Observation: $\phi^*(X)$ is dense in X^* since any open neighborhood of ∞ is of the form $X^* \setminus C$ for some compact $C \subseteq X$, and since X is non-compact, $X \setminus C \neq \emptyset$ so this neighborhood intersects $\phi^*(X)$.

Proposition 14.41. *X^* is compact.*

Proof. Let $\mathcal{U}$ be an open cover of X^*. Since $\infty \in X^*$, $\exists U \in \mathcal{U}$ such that $\infty \in U$. By definition of the topology on X^*, $U = X^* \setminus C$ for some compact $C \subseteq X$. Note that $\mathcal{U}$ covers C, hence $\exists$ finite $\mathcal{U}^\circ \subseteq \mathcal{U}$ that covers C. Therefore, $\mathcal{U}^\circ \cup \{U\}$ is a finite subcover of X^*. □

Proposition 14.42. *X^* is Hausdorff.*

Proof. Let $x, y \in X^*$ be distinct.

- Case 1: $x, y \in X$. Since X is Hausdorff, $\exists$ open $U, V \subseteq X$ such that $x \in U, y \in V$ and $U \cap V = \emptyset$. Note that U, V are also open in X^* and separate x and y.
- Case 2: WLOG let $x \in X$ and $y = \infty$. Since X is locally compact, $\exists$ compact neighborhood C of x in X. Note that $U = C^\circ$ is an open neighborhood of x in X^* and $V = X^* \setminus C$ is an open neighborhood of ∞ in X^*. Note that $U \cap V = \emptyset$.

□

Exercise 14.43. Show that $\mathcal{T}^*$ is a topology on X^*.

Theorem 14.44. *The one-point compactification is unique up to homeomorphism. Let X be a non-compact, locally compact, Hausdorff space. Let $(\tilde{X}, \tilde{\phi})$ be a compactification of X.*

- *$\exists$ quotient map $h : \tilde{X} \to X^*$ such that $h \circ \tilde{\phi} = \phi^*$*
- *Furthermore, if $(\tilde{X}, \tilde{\phi})$ is a one point compactification then h can be chosen to be a homeomorphism.*

$$\begin{array}{ccc} X & \xrightarrow{\widetilde{\phi}} & \widetilde{X} \\ & \searrow^{\phi^*} & \downarrow h \\ & & X^* \end{array}$$

sketch of proof. Define $h : \widetilde{X} \to X^*$ by

$$h(y) = \begin{cases} \phi^*\big((\widetilde{\phi})^{-1}(y)\big), & \text{if } y \in \widetilde{\phi}(X), \\ \infty, & \text{if } y \in \widetilde{X} \setminus \widetilde{\phi}(X). \end{cases}$$

□

Remark 14.45. X is a topological space. if I can find and embedding $\psi \colon X \to [0,1]^J$ such that $\beta X = \overline{\psi(X)} \subseteq [0,1]^J$ is compact, then $(\beta X, \psi)$ is called the *Stone-Cech compactification* of X.

Theorem 14.46. *A necessary and sufficient condition for the existence of such and embedding is that X is completely regular.*

Definition 14.47. A topological space X is *completely regular* if X is Hausdorff and for every point p and each closed set $C \not\ni p$ can be separated by a continuous function, i.e., $\exists f \colon X \to [0,1]$ continuous such that $f(p) = 0$ and $f(C) = \{1\}$.

$$\begin{array}{ccc} X & \xrightarrow{\psi} & \beta X \\ & \searrow^{f} & \downarrow h \\ & & Y \end{array}$$

Example 14.48. .

- $\mathbb{R} \to \beta\mathbb{R} \cong [-\infty, \infty] \cong [0, 1]$
- $\mathbb{N} \to \beta\mathbb{N} \cong \{1/n : n \in \mathbb{N}\} \cup \{0\} \subset [0, 1]$

Theorem 14.49. *Any countable product of sequentially compact spaces is sequentially compact.*

$$X_1, X_2, X_3, \dots \text{ sequentially compact } \implies \prod_{n=1}^{\infty} X_n \text{ sequentially compact}$$

Proof. Let $x^{(1)}, x^{(2)}, \dots$ be a sequence in

$$X = \prod_{k=1}^{\infty} X_k.$$

Then each term $x^{(n)}$ can be written as

$$\begin{aligned} x^{(1)} &= (x_1^{(1)}, x_2^{(1)}, x_3^{(1)}, \dots), \\ x^{(2)} &= (x_1^{(2)}, x_2^{(2)}, x_3^{(2)}, \dots), \\ x^{(3)} &= (x_1^{(3)}, x_2^{(3)}, x_3^{(3)}, \dots), \\ &\vdots \end{aligned}$$

Since X_1 is sequentially compact, $\exists$ finite $I_1 \subseteq \mathbb{N}$ such that $x_1^{(n)} \to y_1$ for some $y_1 \in X_1$. Similarly, since X_2 is sequentially compact, $\exists$ finite $I_2 \subseteq I_1$ such that $x_2^{(n)} \to y_2$ for some $y_2 \in X_2$. Continuing in this manner, we obtain a decreasing sequence of infinite sets $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ such that for each k, the sequence $(x_k^{(n)})_{n \in I_k}$ converges to some $y_k \in X_k$. Let $I = \{n_1 < n_2 < n_3 < \dots\}$ where $n_k = min I_k \setminus \{n_1, n_2, \dots, n_{k-1}\}$. We observe that $\forall k \in \mathbb{N}$, $I \setminus I_k$ is finite $\implies \forall k x_k^{(n)} \to y_k$ as $n \to \infty, n \in I$. □

Chapter 15

Separability and Countability Conditions

15.1 Separation Properties

A space is:

- T_2Hausdorff iff $\forall x, y \in X, x \neq y \exists$ open $A \ni x, B \ni y$ such that $A \cap B = \emptyset$
- T_3 Regular iff
  (1) $\forall x \in X \forall$ closed $C \subseteq X, x \notin C \quad \exists$ open $A \ni xB \subseteq C, A \cap B \neq \emptyset$
  (2) Singletons are closed
- T_4 Normal iff
  (1) $\forall$ closed $C, D \subseteq X, C \cap D = \emptyset \quad \exists$ open sets$A \supseteq C, B \supseteq D, A \cap B = \emptyset$
  (2) Singletons are closed
- $T_{31/2}$ Completely Regular iff
  (1) $\forall x \in X \forall$ closed $C \subseteq X, x \notin C \exists$ continuous $f : X \to [0, 1]$ such that $f(x) = 0, f(y) = 1 \quad \forall y \in C$
  (2) Singletons are closed

Remark 15.1. $T_4 \implies T_{31/2} \implies T_3 \implies T_2$

Example 15.2.
- $\mathbb{R}$ is normal
- Every Compact Hausdorff is Normal
- $(0.1)^{\mathbb{R}}$ Regular but not Normal

15.2 Countability Properties

A space is:

- First Countable: if every point has a countable neighbourhood basis.

- Second Countable: if it has a countable basis,
- Separable: if it has a countable dense set.

Example 15.3.
- $\mathbb{R}$ is Second Countable and Separable.
- Every metrizable space is First Countable.
- $\mathbb{R}/\mathbb{Z}$ is not First Countable.
- $\{0,1\}^{\mathbb{R}}$ is not First Countable
- $\mathbb{R}$ with the discrete topology is First Countable but not Second Countable.
- X compact metric $\implies$ $C(X,\mathbb{R})$ is a Banach Space and is Separable with respect to the uniform topology.

Remark 15.4. A **Banach space** is a normed vector space $(X, \|\cdot\|)$ over $\mathbb{R}$ or $\mathbb{C}$ that is *complete* with respect to the metric induced by the norm, that is, every Cauchy sequence in X converges to an element in X.

	Subspace	**Product**	**Quotient**
Hausdorff	:)	:)	:(
Regular	:)	:)	:(
Normal	:(	:(	:(
First Countable	:)	:) (countable) :((uncountable)	:(
Second Countable	:)	:) (countable) :((uncountable)	:(
Separable	:(	:) (countable) :((uncountable)	:)

15.3 Some Implications

- Prop. X metrizable $\implies$ X First Countable
- Prop. X metrizable and seperable $\implies$ X Second Countable Take for example, $D \subseteq X$ countably dense. $\mathcal{B} = \{B_\epsilon x : x \in D, \quad \epsilon \in \mathbb{Q}^+\}$
- Prop. X Second Countable $\implies$ X separable.
- Prop. X Comapact and Hausdorff $\implies$ X is normal.
- Prop. X metrizable $\implies$ X is normal.
- Prop. X is regular and Second Countable $\implies$ X is normal

Proof of 5. Fix a metric d generating the topology of X. Let $P, Q \subseteq X$ be disjoint closed sets. If $P = \emptyset$ or $Q = \emptyset$, then we can choose X and $\emptyset$. Assume $P \neq \emptyset$ and $Q \neq \emptyset$. For every $X \in P$, choose ϵ_x such that $B_{\epsilon_x} \cap Q = \emptyset$, and for every $y \in Q$, choose δ_y such that $B_{\delta_y} \cap P = \emptyset$. Note that: $\forall x \in P. \forall y \in Q, B_{\epsilon_x/3}(x) \cap B_{\delta_y/3}(y) = \emptyset$ Because if $z \in B_{\epsilon_x/3}(x) \cap B_{\delta_y/3}(y)$, then $d(x,z) < \epsilon_x/3$ and $d(y,x) < \delta_y/3 \leq \epsilon_x/3$ and assuming $\epsilon_x \geq \delta_y \implies d(x,y) < 2\epsilon_x/3$ contradiction. Take $U = \bigcup_{x\in A} B_{\epsilon_{x/3}}(x)$ and $W = \bigcup_{y\in B} B_{\delta_{y/3}}(y)$ □

Proof of 6. $A, B \subseteq X$ closed and $A \cap B = \emptyset$. We can assume $A \neq \emptyset$ and $B \neq \emptyset$. Let $\mathcal{B}$ be a countable basis for X. By regularity $\forall x \in A \exists$ open $U_x \ni x$ and open $W_x \supseteq B$ such that $U_x \cap W_x = \emptyset$. Since $\mathcal{B}$ is a basis, we can choose $F_x \in B$ such that $x \in F_x \subseteq U_x$ **Note:** $\overline{F_x} \cap W_x = \emptyset$, hence $\overline{F_x} \cap B = \emptyset$. As we vary x in A, F_x varies over a subset $\mathcal{F} \supset \mathcal{B}$. Hence:

- $\exists F_1, F_2, \cdots \in \mathcal{B}$ such that $\bigcup_{n=1}^{\infty} \supseteq A$ and $\forall n \overline{F_n} \cap B = \emptyset$
- $\exists G_1, G_2, \cdots \in \mathcal{B}$ such that $\bigcup_{n=1}^{\infty} G_n \supseteq B$ and $\forall n \overline{G_n} \cap A = \emptyset$

Recursively define:

$$F_1' = F_1 \setminus \overline{G_1}$$
$$G_1' = G_1 \setminus \overline{F_1}$$
$$F_n' = F_n \setminus \bigcup_{i=1}^{n} \overline{G_i}$$
$$G_n' = G_n \setminus \bigcup_{i=1}^{n} \overline{F_i}$$

Let $\tilde{A} = \bigcup_{n=1}^{\infty} F_n'$ and $\tilde{B} = \bigcup_{n=1}^{\infty} G_n'$. Note that $\tilde{A}$ and $\tilde{B}$ are open, $\tilde{A} \supseteq A$, $\tilde{B} \supseteq B$ and $\tilde{A} \cap \tilde{B} = \emptyset$. □

15.4 Urysohn's Lemma

Let X be normal. Let $A, B \subseteq X$ be disjoint closed sets. Then, $\exists f \colon X \to [0,1]$ such that $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$.

Proof of Urysohn's Lemma when X is metrizable. Let d be a metric consistent with X. Define $f(x) = \frac{d(x,A)}{d(x,A)+d(x,B)}$ where $d(x, A) = \inf\{d(x, a) : a \in A\}$ and $d(x, B) = \inf\{d(x, b) : b \in B\}$. Note that $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$. □

Lemma 15.5. *X is normal iff for every closed sets $A, B, \subset X$ there exists open $U \subseteq X, A \subset U$ and $\overline{U} \cap B = \emptyset$*

Proof of Urysohn's lemma when X is normal. LeT $U_{1/2}$ be an open set such that $A \subseteq U_{1/2}$ and $\overline{U_{1/2}} \cap B = \emptyset$. Recursively define for each $n \in \mathbb{N}$ and $0 \leq k \leq 2^n$ open sets $U_{k/2^n}$ such that:

- $U_0 = X \setminus B$
- $U_1 = X \setminus A$
- $U_{k/2^n} \subseteq U_{(k+1)/2^n}$
- $\overline{U_{k/2^n}} \subseteq U_{(k+1)/2^n}$

Note: the indices are the dyadic rationals in $[0, 1]$ and are dense in $[0, 1]$. $\mathbb{Q}_2 = \{r = k/2^n : n \in \mathbb{N}, 0 \leq k \leq 2^n\} \subseteq \mathbb{Q}$ and they are dense in $\mathbb{R}$. In summary, for each $r \in \mathbb{Q}_2$, $0 < r < 1$, we have open sets $U_r \subseteq X$ such that:

- $A \subseteq U_r$ for all $r > 0$
- $\overline{U_r} \cap B = \emptyset$ for all $r < 1$
- $r < s \implies U_r \subseteq U_s$

Define $f\colon X \to [0,1]$ by

$$f(x) = \begin{cases} \inf\{r : x \in U_r\} & x \in \bigcup_{0<r<1} U_r \\ 1 & \text{otherwise} \end{cases}$$

Observe that: $x \in A \implies f(x) = 0$ and $x \in B \implies f(x) = 1$. More observations:

- if $x \in U_r$ then $f(x) \leq r$
- if $x \notin U_r$ then $f(x) \geq r$
- if $x \in \overline{U_r}$ we still have $f(x) \leq r$

Claim. f is continuous.

Proof. Let $x \in X$ and $\epsilon > 0$.

- Case 1: $0 < f(x) < 1$. Let $r, s \in \mathbb{Q}_2$ such that $f(x) - \epsilon < r < f(x) < s < f(x) + \epsilon$ and $0 < r < s < 1$. $x \in U_s$ and $x \notin \overline{U_r}$, $U_s \setminus \overline{U_r}$ is open, and $\forall y \in U_s \setminus \overline{U_r}, f(x) - \epsilon < r < f(y) < s < f(x) + \epsilon$. Therefore, f is continuous at x.
- Case 2: $f(x) = 0$. Let $s \in \mathbb{Q}_2$ such that $0 < s < \epsilon$. Note that $x \in U_s$. For all $y \in U_s, f(y) < s < \epsilon$. Therefore, f is continuous at x.
- Case 3: $f(x) = 1$. Let $r \in \mathbb{Q}_2$ such that $1 - \epsilon < r < 1$. Note that $x \notin \overline{U_r}$. For all $y \notin \overline{U_r}, f(y) > r > 1 - \epsilon$. Therefore, f is continuous at x.

□

□

15.5 Tietze Extension Theorem

Let X be normal. Let $A \subseteq X$ be closed then every continuous function $f\colon A \to \mathbb{R}$ can be extended to a continuous function $\tilde{f}\colon X \to \mathbb{R}$ such that $\tilde{f}|_A = f$.

Proof. Proof is trivial left as an exercise □

15.6 Embedding into cubes, metrization, compactification

Definition 15.6. Let $a \in X$ and $a \notin B \subseteq X$ with B closed. A continuous map $f\colon X \to \mathbb{R}$ *separates* a from B means $f(a) \notin \overline{f(B)}$ A family $\mathcal{F}$ of continuous functions separates points from closed sets if $\forall a \in X$ and $\forall B \subseteq X$ closed with $a \notin B, \exists f \in \mathcal{F}$ that separates a from B.

Remark 15.7. Note that if X is completely regular, then $C(X, [0,1])$ separates points from closed sets.

Theorem 15.8. *Let X be a space in which the singletons are closed. Let $\mathcal{F} = \{f_j \in C(X, [0,1]) : j \in J\}$ be a family of continuous functions that separates points from closed sets. Define $\psi\colon X \to [0,1]^J$ by $\psi(x) = (f_j(x))_{j \in J}$.*

Proposition 15.9. *Suppose X is regular and second countable. Then, there exists a countable family $\mathcal{F} \subseteq C(X, [0,1])$ that separates points from closed sets.*

Corollary 15.10 (Urysohn Metrization Theorem)**.** *Every regular, second countable space is metrizable.*

Proof of theorem. We need to show:

(1) ψ is injective.

(2) ψ is continuous.

(3) $\psi^{-1}\colon \psi(X) \to X$ is continuous.

Let's prove each of these statements:

(1) let $x, y \in X$, $x \neq y$, Pick j such that f_j separates x and $\{y\}$., then $F_j(x) \neq f_j(\{y\})$, hence $f_j(x) \neq f_j(y)$ therefore $\psi(x) \neq \psi(y)$ so ψ is injective.

(2) ψ is coordinate wise continuous hence continuous.

(3) Let $U \subseteq X$ be open. We need to show that $\psi(U)$ is open in $\psi(X)$. Let $y \in \psi(U)$, so $\exists x \in U$ such that $\psi(x) = y$. Since $X \setminus U$ is closed and $x \notin X \setminus U$, $\exists j \in J$ such that f_j separates x from $X \setminus U$. Let $\epsilon > 0$ such that $f_j(x) \pm \epsilon \cap \overline{f_j(X \setminus U)} = \emptyset$. Since f_j is continuous, we can find an open subset $V \ni x$ such that $f_j(V) \subseteq f_j(x) \pm \epsilon$. Let $W = \pi_j^{-1}(f_j(x) \pm \epsilon) = \{u \in [0,1]^J : u_j \in f_j(x) \pm \epsilon\}$. Note:

- W is Open
- $W \cap \psi(X)$ is open in $\psi(X)$
- $W \cap \psi(X) \subseteq \psi(U)$ since if $z \in W \cap \psi(X)$, then $\exists z' \in X$ such that $\psi(z') = z$ and $f_j(z') \in f_j(x) \pm \epsilon$ so $z' \notin X \setminus U$ hence $z' \in U$ so $z \in \psi(U)$.
- $w \in W \cap \psi(x)$

Therefore, $\psi(U)$ is open in $\psi(X)$.

□

Proof of proposition. Note: X is normal. By Urysohn's Lemma, for every $p \in X$ and every closed $Q \subseteq X$ with $p \notin Q$, we can find $f \in C(X, [0,1])$ that separates p from Q. Let $\mathcal{B}$ be a countable basis. For $A, B \in \mathcal{B}$ with $\overline{A} \subseteq B$, let $f_{A,B}\colon X \to [0,1]$ be continuous such that

$$\begin{cases} f_{A,B}(x) = 0 & \text{if } x \in \overline{A}, \\ f_{A,B}(x) = 1 & \text{if } x \notin B. \end{cases}$$

Let $J = \{(A, B) \in \mathcal{B}^2 : \overline{A} \subseteq B\}$.

Claim. $\mathcal{F} = \{f_{A,B} : (A, B) \in J\}$ separates points from closed sets.

Proof. Let $p \in X$ and $Q \subseteq X$ be closed, with $p \notin Q$. Since $X \setminus Q$ is open and $\mathcal{B}$ is a basis, there exists $B \in \mathcal{B}$ such that $p \in B \subseteq X \setminus Q$. By regularity, since $X \setminus B$ is closed and $p \notin X \setminus B$, we can find an open set $U \ni p$ such that $\overline{U} \subseteq B$. Since $\mathcal{B}$ is a basis and U is open, there exists $A \in \mathcal{B}$ such that $p \in A \subseteq U$. Hence $p \in A \subseteq \overline{A} \subseteq B$. We have $f_{A,B}(p) = 0$ because $p \in \overline{A}$, and $f_{A,B}(x) = 1$ for all $x \in Q$. □

□